

A New Analysis of the Morphosyntactic–Semantic Structure of Japanese Verbal Adjectives — *KU*-adjectives¹

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Abstract:

The present paper reanalyses the morphosyntactic–semantic structure of Japanese verbal adjectives (VAs), also referred to as *i*-adjectives. Traditionally, inflected forms of this class, such as *takaku* from *takai* 'be high/tall', have been analysed as consisting of the stem such as *taka-* and the ending *-ku*. However, given the *multiple* morphosyntactic and semantic parallelisms with verbal stems, it is more plausible that *takaku* as a whole functions as the stem of the verbal adjective.

This analysis thus analyses *taka-* as one of the two immediate constituents of the VA stem *takaku*, which encodes, as a VA radical, the lexical meaning specific to the VA, and *-ku* as the other constituent, functioning as the VA thematic — namely, stem-forming — suffix, which licences the VA radical as a syntactic constituent of the VA category at the sentential level.

The discussion then examines how VAs encode the tense value [–PAST], proposing two alternative interpretations: either the phonologically null inflexional suffix $-\emptyset$, as in [[[*taka*]*i*] $\emptyset$], acquires the [–PAST] value through paradigmatic opposition to *-ta*, specified as [+PAST], as in [[[*taka*]*k+at*]*ta*]; or *-i*, as in [[[*taka*]*i*], is an amalgam of the VA thematic suffix and the non-past tense suffix. On the other hand, the atemporal form *takaku* lacks any temporal suffix, as in [[[*taka*]*ku*]. Based on this, the paper further examines the allomorphy of the VA thematic suffix: $-ku \approx -i \approx -(k)i \approx -ke$.

The study also examines the structural parallels between VAs and true verbs and the lack of surface-level parallelism between the negative forms of VAs and those of true verbs. Finally, it surveys syntactic and logical-propositional differences between verbal-adjectival and nominal-adjectival constructions.

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1. Introduction

The inflected forms of Japanese verbal adjectives (VAs) — simply adjectives, canonical adjectives, or *i*-adjectives — are not so transparent in their morphosyntactic structure as those of verbs. In verbs, the boundary between the stem and the ending can be clearly determined. Taking the verb *taberu* 'eat' as an example, the non-past form ***taberu*** consists of the stem *tabe-* and the inflexional ending consisting of the tense suffix *-ru*, specified as [–PAST], while the past form ***tabeta*** consists of the same stem *tabe-* and the inflexional ending composed of the tense suffix *-ta*, specified as [+PAST]. In other words, the opposition between the two forms is reduced to a paradigmatically transparent opposition between the suffixes *-ru* and *-ta*, both sharing the same stem *tabe-* as their inflexional base.

With VAs, on the other hand, taking *takai* 'be high/tall' as an example, the non-past form is ***takai***, whereas the past form is ***takakatta***. Their relationship is not as straightforward as the opposition between the non-past and past forms of verbs such as *taberu* and *tabeta*, and the boundary between the stem and the ending is, in our view, not easy to determine. Tsujimura (2007: 119) states that "adjectives are identified by a variety of conjugational endings, such as the non-past ending *-i* ... and the past tense ending *-kat-ta*." Following this, in her analysis of the VA *takai*, she treats the non-past form *takai* as consisting of the stem *taka-* + the ending *-i*, and the past form *takakatta* as the stem *taka-* + the ending *-kat-ta* (henceforth ' + ' with spaces on both sides indicates concatenation). This appears to represent one of the traditional and standard analyses in Japanese linguistics.

With VAs, on the other hand, taking *takai* 'be high/tall' as an example, the non-past form is ***takai***, whereas the past form is ***takakatta***. Their relationship is not as straightforward as the opposition between the non-past and past forms of true verbs such as *taberu* and *tabeta*, and the boundary between the stem and the ending is, in our view, not easy to determine. Tsujimura (2007: 119) states that "adjectives are identified by a variety of conjugational endings, such as the non-past ending *-i*, and the past tense ending *-kat-ta*." On this basis, in her analysis of the VA *takai*, she treats the non-past form *takai* as consisting of the stem *taka-* + the ending *-i*, and the past form *takakatta* as the stem *taka-* + the ending *-kat-ta* (henceforth, ' + ', with spaces on both sides, indicates concatenation). This appears to represent a standard traditional analysis in Japanese linguistics.

This paper, in contrast, proposes a different type of analysis, assuming that the formation of the basic stem of verbal adjectives — the VA stem — takes place at the level of the lexicon, while the combination of that basic stem with the inflexional ending occurs at the level of syntax, since such endings are assumed to constitute phrasal heads within the sentence structure. Based on this

assumption, the morphosyntactic structure of verbal adjectives is analysed as in (1), where the symbols '-' (suffixation), '()' (deletion) and '+' with no spaces on either side (compounding), when attached immediately to morphemes, indicate the respective processes:

(1)

Word			
Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i>	—	-Ø [-PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-k(u)</i>	+at-	-ta [+PAST]
<i>taka-</i>	<i>-ke</i>	—	-reba 'if (cond.)'

As mentioned above, the conclusive form *takai* — the so-called *syūsikei* in Japanese — is traditionally analysed as the stem *taka-* + the ending *-i*, but here, *takai* as a whole is treated as the lexical-level stem — that is, the VA stem. The root *taka-* is then regarded as the VA radical, which encodes the lexical meaning specific to the VA, while *-i* is analysed as the VA thematic suffix — namely, the VA stem-forming suffix. To this VA stem *takai*, the phonologically null inflexional ending *-Ø*, specified as [-PAST], is attached as in *takai-Ø*.

Similarly, the conjunctive/adverbial form *takaku* — the so-called *ren'yōkei* — is generally analysed as the stem *taka-* + the ending *-ku*. Here, however, *takaku* as a whole is likewise treated as the VA stem, with *taka-* regarded as the VA radical and *-ku* as the default/elsewhere allomorph of the VA thematic suffix. On the other hand, unlike the above *takai-Ø*, with the ending *-Ø* specified as [-PAST], *takaku* lacks any inflexional ending and is thus analysed as consisting solely of the stem, given its atemporality, much like the conjunctive form *tabe* from the verb *taberu*.

In the past form *takak+at-ta* — which is generally analysed as the stem *taka-* + the complex ending *-kat-ta* — the string *takak*, derived from *takaku* by final-vowel deletion, is considered the lexical-level VA stem: when the VA stem *takaku* combines with the phrasal-head tense suffix *-ta*, specified as [+PAST], at the sentential level, the stem *ar-* from the light verb *ar-u* 'be' is inserted immediately after it, as in *takaku#ar-ta*, owing to the incompatibility of the VA stem *takaku* with the verbal suffix *-ta*. This yields the final form *takak+at-ta*, through the deletion of the stem-final *-u* in *takaku* before *ar-* — which indicates the compounding of *takaku* with *ar-* — and the assimilation *ar- /ar-/* → *at- [at-]* before *-ta*.

Finally, the conditional form *takake-reba* — the so-called *kateikei* — is generally analysed as the stem *taka-* + the ending *-kereba*. Here, however, *takake* as a whole is treated as the VA stem, with *taka-* regarded as the VA radical and *-ke* as another allomorph of the VA thematic suffix, while *-reba* is exactly the same inflexional suffix found in the verbal conditional form *tabe-reba* from

tabe-ru. The following discussion examines the justification for the analysis presented in (1). The key concepts are as follows:

- **Multiple parallels — morphosyntactic and semantic — between true-verbal stems and the conjunctive/adverbial forms of VAs**
- **VA stem consisting of "VA radical + VA thematic suffix: *-ku* $\approx$ *-i* $\approx$ *-(k)i* $\approx$ *-ke*"**
- **Obligatorily inserted stem *ar-* $\approx$ *at-* $\approx$ *ara-* of the light verb *ar-u* 'be' in certain constructions**
- **Phonologically null tense suffix *-Ø*, specified as [–PAST]**
- **Suppletive replacement of **ara-nai* 'not be' by *nai***
- **Syntactic and logical-propositional contrast between *ku*-adjectives and *ni*-adjectives**

2. The identification of the stem of verbal adjectives

Traditionally, as repeatedly noted above, the conclusive form of VAs has been analysed as consisting of a stem followed by the inflexional ending *-i* — e.g. *takai* consisting of *taka-* + *-i*. On the other hand, the conclusive form of verbs — particularly vowel-stem verbs — has been analysed as consisting of a stem followed by the inflexional ending *-ru*. In *taberu*, for instance, *tabe-* is considered the stem and *-ru* the ending. **Nevertheless, although both *taka-* and *tabe-* have been treated as stems, their syntactic behaviour differs fundamentally.** The verbal stem *tabe*, without an overt ending, can appear on its own within a sentence, as in (2), whereas the so-called adjectival stem *taka-* can never occur independently in such contexts, as shown in (3), where AGT, THEM, POSS, TOP, ACC, NOM, CONJ, PST and NPST stand for *agent, theme, possessor, topic, accusative, nominative, conjunctive, past* and *non-past*, respectively:

(2) *kare=wa gohan=o tabe (sosite) de-te it-ta / ik-u*
 he(AGT)=TOP meal(THEM)=ACC eat (CONJ) leave-CONJ go-PST / go-NPST
 'He, having a meal, (then) left / leaves.'²

(3) *kanozyo=wa se=ga *taka- (sosite) utukusikat-ta / utukusii*
 she(POSS)=TOP stature(THEM)=NOM be_high (CONJ) be_beautiful-PST / be_beautiful-NPST
 'She, being tall, was / is (also) beautiful.' (ungrammatical)

² The translations of the examples in this paper are *deliberately literal* and may therefore sound unnatural — sometimes very unnatural — in English.

What can occur in the same syntactic context as the verbal stem *tabe* is not the so-called adjectival stem *taka-* itself, but rather the conjunctive/adverbial form *takaku*, as in (4), which has traditionally been analysed — like *takai* — as consisting of the so-called stem *taka-* and the so-called inflexional ending *-ku*:

- (4) *kanozyo=wa se=ga takaku (sosite) utukusikat-ta / utukusii*
 she(POSS)=TOP stature(THEM)=NOM be_high (CONJ) be_beautiful-PST / be_beautiful-NPST
 'She, being tall, was / is (also) beautiful.' (grammatical)

Accordingly, the verbal stem *tabe* and the adjectival constituent *takaku* share the same syntactic property of being **able to occur on their own at the end of non-sentence-final coordinated clauses**. Moreover, as is clear from (2) and (4), at the point in the utterance represented by *kare=wa gohan=o tabe* and *kanozyo=wa se=ga takaku* in the respective syntactic strings, **it remains indeterminate whether the utterance refers to a past or non-past situation**. Temporal specification is only achieved by the sentence-final predicate — *it-ta / utukusikat-ta* in the past and *ik-u / utukusii* in the non-past. Therefore, the verbal stem *tabe* and the VA constituent *takaku* share the same semantic property of being **atemporal**: they do not themselves make any temporal reference, indicating that they both **lack inflexional tense endings** — a **defining feature of the predicative stem**. Furthermore, the similarity between the verbal stem *tabe* and the VA constituent *takaku* is not limited to these points; both **can appear within sentences followed by the conjunctive inflexional suffixes *-te* 'and', *-temo* 'even though; even if' and *-tatte* 'even if'**, as shown in (5)–(10), where CONC.CONJ, CONC.COND.CONJ and NEG stand for *concessive conjunctive*, *concessive conditional conjunctive* and *negative*, respectively:

- (5) *kare=wa gohan=o tabe-te de-te it-ta / ik-u*
 he(AGT)=TOP meal(THEM)=ACC eat-CONJ leave-CONJ go-PST / go-NPST
 'He, having a meal, then left / leaves.'
- (6) *kanozyo=wa se=ga takaku-te utukusikat-ta / utukusii*
 she(POSS)=TOP stature(THEM)=NOM be_high-CONJ be_beautiful-PST / be_beautiful-NPST
 'She, being tall, was / is also beautiful.'
- (7) *kare=wa gohan=o tabe-temo hutora- -nai*
 he(AGT)=TOP meal(THEM)=ACC eat-CONC.CONJ gain_weight NEG-NPST
 'He, even having a meal, does not gain weight.'
- (8) *kanozyo=wa nedan=ga takaku-temo sore=o ka-u*
 she(AGT)=TOP price(THEM)=NOM be_high-CONC.CONJ it(THEM)=ACC buy-NPST
 'She, even the price being high, buys it.'
- (9) *kare=wa gohan=o tabe-tatte hutora- -nai*
 he(AGT)=TOP meal(THEM)=ACC eat-CONC.COND.CONJ gain_weight NEG-NPST
 'He, even if he has a meal, does not gain weight.'

- (10) *kanoz yo=wa nedan=ga takaku-tatte sore=o ka-u*
 she(AGT)=TOP price(THEM)=NOM be_high-CONC.COND.CONJ it(THEM)=ACC buy-NPST
 'She, even if the price is high, buys it.'

In contrast, the forms **taka-te*, **taka-temo* and **taka-tatte* — each composed of the so-called stem *taka-* followed respectively by the inflexional suffixes *-te*, *-temo* and *-tatte* — are all ungrammatical. In this way, there is clear syntactic, semantic and morphosyntactic parallelism between *tabe(-)* and *takaku(-)*. Given that *tabe(-)* is analysed as the stem of the verb, it seems more appropriate to treat *takaku(-)* — not as the concatenation of the so-called stem *taka-* and the so-called inflexional ending *-ku*, as traditionally assumed — but rather as the stem of the verbal adjective in its entirety, which we shall henceforth refer to as the VA stem.

Historically, there was an etymological form in Early Middle Japanese from which the modern verbal form *tabe-tatte* originated — namely, *tabe-tari-tote*, which later changed to *tabe-ta-tote*. In contrast, the modern VA form *takaku-tatte* does not have a comparable direct etymon with the same structure in Early Middle Japanese, such as **takaku-tari-tote*, nor in later Japanese, such as **takaku-ta-tote* — both of which would have been ungrammatical. This suggests that the modern grammatical form *takaku-tatte* was formed by inserting the VA stem *takaku-* into the "verbal stem" position within the already existing template "verbal stem + inflexional suffix *-tatte*" — as in *tabe-tatte* — on the basis of their morphosyntactic–semantic commonality as predicative stems. If *taka-* were, like *tabe-*, a predicative stem, it would be impossible to explain why **taka-tatte* — obtained by replacing the genuine stem *tabe-* with the so-called stem *taka-* in *tabe-tatte* — results in ungrammaticality. In other words, the very existence of such inflected forms as *takaku-tatte* can be taken as evidence that the VA stem, which has the same predicative-stem status as verbal stems such as *tabe(-)*, is not *taka-* but *takaku(-)*. All of the above discussion leads to the following question: how, then, should *taka-* — traditionally regarded as the stem — and *-ku* — traditionally regarded as the inflexional ending — be reanalysed as morphological constituents? This issue will be discussed in the following section.

3. The radical and the thematic suffix of verbal adjectives

When we observe all the inflected forms of the VA *takai* — namely, *takai* $\approx$ *taka(k)i* $\approx$ *takaku* $\approx$ *takakute* $\approx$ *takakutemo* $\approx$ *takakutatte* $\approx$ *takakatta* $\approx$ *takakattarō* $\approx$ *takakattara(ba)* $\approx$ *takakattari* $\approx$ *takakarō* $\approx$ *takakare* $\approx$ *takakereba* — we find that they all share the element *taka-*. This *taka-* is also attested in derived nouns such as *takasa* 'height' and *takami* 'loftiness' as well as in compound nouns such as *takanozomi* 'high hope' and *takadai* 'high platform', where it refers to the very property or state of 'being high/tall'. Since it cannot be further decomposed into smaller

morphemes, it is to be regarded as a root. However, as the ungrammaticality of (11) clearly shows, the root *taka-* alone cannot function as a verbal adjective — that is, it cannot be inserted into a terminal node directly dominated by the VA node — and thus cannot establish structural relations with other constituents within the sentence (SRC and POST in the glosses below stand for *source* and *postposition*, respectively):

- (11) *koko=kara *taka kyodaina tō=ga mie-ru*
 here(SRC)=POST be_tall be_colossal-NPST tower(THEM)=NOM be_seen-NPST
 'From here, a tower that, being tall, is colossal can be seen.' (ungrammatical)

By contrast, example (12), in which *taka-* is accompanied by *-ku*, is grammatical:

- (12) *koko=kara takaku kyodaina tō=ga mie-ru*
 here(SRC)=POST be_tall be_colossal-NPST tower(THEM)=NOM be_seen-NPST
 'From here, a tower that, being tall, is colossal can be seen.' (grammatical)

Now, as the derived words and compound words cited above show, **the root *taka-* functions only as a mere constituent of word formation**: it cannot itself serve as a syntactic unit at the sentential level. We are thus led to posit that *taka-* requires an additional element in order to function as the syntactic category VA at the sentential level. This element is precisely *-ku*, which is present in the sentential-level form *takaku*, but absent from word-level forms such as *takasa*, *takami*, *takanozomi* and *takadai*. As observed in Section 2, the addition of *-ku* to the root *taka-* yields a verbal adjective stem that is functionally — more precisely, morphosyntactically and semantically — equivalent to the verbal stem *tabe-*. We shall therefore refer to *-ku* as the **VA thematic suffix** — namely, the **VA stem-forming suffix**. (Note that from the perspective of Distributed Morphology, this constituent may be regarded as an *overt* instantiation of the categorial head *a*, a category-defining element that is otherwise often assumed to be *covert*.)

The historical development of verbal adjectives also supports this interpretation: "Originally, adjectives, in their stem form, were placed before nouns and verbs and modified them, as in *nagauta* and *takayuku*. However, in order to express explicitly the functions of the conjunctive, adnominal, conclusive forms, *-ku*, *-ki*, *-si* and the like were added, as in *takaku*, *takaki*, *takasi*. ... After the Nara period, realis forms (*izenkei*) such as *takakere* developed. In Japanese, since adjectival sentences were used less frequently than verbal sentences, the forms of adjectives serving as predicates developed more slowly..." (Ōno, S., et al. (Eds.). 1990: 1467, translated from Japanese). From the perspective of this paper, **the root of VAs** — e.g. *naga-* 'long' and *taka-* 'high/tall' — **originally functioned only as a formative element at the lexicon level, forming compounds** such as *naga+uta* 'long poem' and *taka+yuku* 'fly high into the sky', **placed before base nouns or base verbs. However, in order to function as the head of a VA phrase at the**

sentential level, that is, to function as the category VA, the thematic suffix *-ku* $\approx$ *-ki* $\approx$ *-si* $\approx$ *-ke*, as in *taka-ku* $\approx$ *taka-ki* $\approx$ *taka-si* $\approx$ *taka-ke* — corresponding to *-ku* $\approx$ *-(k)i* $\approx$ *-i* $\approx$ *-ke* in Contemporary Japanese, as in *taka-ku* $\approx$ *taka-(k)i* $\approx$ *taka-i* $\approx$ *taka-ke* — was added. Note that the VA thematic suffix proposed in this paper corresponds to what Frellesvig (2010: 80–90, *inter alia*) refers to as the "adjectival copula", although its allomorphs *-ku* $\approx$ *-ki* $\approx$ *-si* $\approx$ *-ke* constitute only a small part of his "Old Japanese adjectival copula forms" (listed in Table 3.8, *ibid.*: 81). The origin of these conjunctive/adverbial form, adnominal form, conclusive form and realis form ending in the VA thematic suffix *-ku* $\approx$ *-ki* $\approx$ *-si* $\approx$ *-ke* — namely, *ren'yōkei*, *rentaikei*, *syūsikei* and *izenkei* — is explained in Okimori (2017: 85–88).

Let us now turn to the morphological status of the constituent *taka-* within the VA stem. In the preceding discussion, *taka-* was referred to as a *root* in relation to the suffix *-ku*, but this designation is not strictly accurate from the viewpoint of morphological structure. As the constituent *na+daka-* — derived from *na* 'name' + *taka-* 'high', with *rendaku* in the second element — in the compound verbal adjective *na+daka-i* 'be famous' in (13) shows, what precedes the VA thematic suffix need not be a single root, but may also involve root–root compounding (VA.TH in the glosses below stands for VA thematic suffix):

- (13) *Yamazaki=wa na+daka-i gengogakusya=de ar-u*
 Yamazaki(THEM)=TOP name+high-VA.TH-NPST linguist=POST-CONJ exist-NPST
 'Yamazaki is a famous linguist.'

It is therefore morphologically inappropriate to refer to the constituent preceding the VA thematic suffix *-ku* $\approx$ *-i* $\approx$ *-(k)i* $\approx$ *-ke* as a root. In what follows, we shall refer to it as a **VA radical**, in the sense of **one of the two immediate constituents of a VA stem that constitutes its semantic core**. As the inflected forms *takaku*, *takaku-te*, *takaku-temo* and *takaku-tatte* in (4), (6), (8) and (10) show, the VA stem formed with the thematic suffix *-ku* — namely, *takaku-* — can appear on its own or followed by the suffixes *-te*, *-temo* and *-tatte* at the sentential level. On this basis, the structure of verbal adjectives may be analysed as in (14):

(14)

Word		
VA stem		Ending
VA radical	VA thematic suffix	Inflexional suffix
<i>taka-</i>	<i>-ku</i>	—
<i>taka-</i>	<i>-ku</i>	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>

It can be confirmed that the morphemes *-te*, *-temo* and *-tatte* are inflexional suffixes, as follows. In *takaku-te*, *takaku-temo* and *takaku-tatte*, the focus particle *=mo* 'too, also' or the topicalisation/

contrast particle =*wa* 'as for; contrastively' — both of which are enclitics — can be syntactically inserted immediately after the VA stem *takaku-*. In these cases, the stem allomorph *at-* — underlyingly, *ar-* — of the light verb *ar-u* 'be' is obligatorily inserted before *-te*, *-temo* and *-tatte*, as shown in (15), where '#' represents a morphological word boundary that coincides with a prosodic word boundary and '=' indicates enclisis:

- (15) *takaku-te* → *takaku=mo#at-te* vs. **takaku=mo#te* 'being also high/tall'
takaku-temo → *takaku=wa#at-temo* vs. **takaku=wa#temo* 'high/tall, even though/if it is so'
takaku-tatte → *takaku=wa#at-tatte* vs. **takaku=wa#tatte* 'high/tall, even if it is so'

In the case of the verbal inflected forms *tabe-te*, *tabe-temo* and *tabe-tatte*, formed with the same suffixes as in (5), (7) and (9), the situation is exactly the same, as indicated in (16):

- (16) *tabe-te* → *tabe=mo#si-te* vs. **tabe=mo#te* 'doing also eat'
tabe-temo → *tabe=wa#si-temo* vs. **tabe=wa#temo* 'even though/if it does eat'
tabe-tatte → *tabe=wa#si-tatte* vs. **tabe=wa#tatte* 'even if it does eat'

In this case, a stem allomorph *si-* of the light verb *su-ru* 'do' is inserted before *-te*, *-temo* and *-tatte*. From (15) and (16), the following observations (17)–(20) may be drawn:

(17) In *takaku-te*, *takaku-temo* and *takaku-tatte* in (15), and *tabe-te*, *tabe-temo* and *tabe-tatte* in (16), respectively, it is possible to insert the enclitic particle =*mo* or =*wa* immediately after the VA stem *takaku-* and the verbal stem — henceforth, the V stem — *tabe-* at the syntactic level. This indicates that *takaku-* and *tabe-* can occupy the same structural position in syntax and, here too, **syntactic parallelism is observed between *takaku-* and *tabe-***.

(18) In both (15) and (16), if *-te*, *-temo* and *-tatte* were free morphemes and could constitute autonomous words, then forms such as **takaku=mo#te*, etc. and **tabe=mo#te*, etc. should be grammatical as they are. This is not the case, however: only when the stems *at-* and *si-* of the light verbs *ar-u* and *su-ru* are inserted, respectively, do the forms become grammatically adequate. It follows, then, that *-te*, *-temo* and *-tatte* are inflexional suffixes which cannot constitute a word independently and, formally, always require a stem.

(19) Based on (18), the insertion of the light verb stems *at-* and *si-* in *takaku-te* → *takaku=mo#at-te*, etc. in (15) and *tabe-te* → *tabe=mo#si-te*, etc. in (16) indicates that, since

both *takaku-* and *tabe-* function as stems for the inflexional suffixes *-te*, *-temo* and *-tatte* in *takaku-te*, etc. and *tabe-te*, etc., the insertion of the enclitic particle *=mo* or *=wa* between the stems and the suffixes renders the latter stemless, and therefore requires the subsequent insertion of alternative formal stems — namely, *at-* and *si-* — before them, because *=mo* and *=wa* cannot function as stems for them. This again supports the claim that **it is not *taka-* but *takaku-* in its entirety that functions as the stem in VAs** and, here too, **morphosyntactic parallelism is observed between *takaku-* and *tabe-***.

(20) In (15), the reason why it is not some VA stem but the V stem *at-* that is inserted in *takaku=mo#at-te*, etc. is simply that Japanese lacks a class of light verbal adjectives and their corresponding stems, and hence some V stem must instead be employed to formally support the suffixes *-te*, *-temo* and *-tatte* even in *takaku=mo#at-te*, *takaku=wa#at-temo* and *takaku=wa#at-tatte*. But why is it not the stem *si-* from *su-ru* but the stem *at-* from *ar-u* that is selected? The motivation seems to be that the prototypical categorial distinction — *stativity* vs. *dynamicity* — between the VA stem *takaku-* and the V stem *tabe-* conditions the choice between the stem of the *stative* light verb *ar-u* 'be' and that of the *dynamic* light verb *su-ru* 'do'. The syntactic process that determines this selection based on semantics remains a topic for future investigation.

Taken together with the description at the end of Section 2, it can be observed that **clear parallelism in multiple respects — morphosyntactic and semantic — is evident between the VA stem *takaku-* and the V stem *tabe-***.

Next, let us turn to more complex structures: in the inflected forms *taka-k+at-ta*, *taka-k+at-tar-ō*, *taka-k+at-tar-a(-ba)*, *taka-k+at-tar-i*, *taka-k+ar-ō* and *taka-k+ar-e*, if the focus particle *=mo* 'too, also' is inserted after *taka-k*, before the inflected forms of the light verb *ar-u* 'be' — namely, *at-ta*, *at-tar-ō*, *at-tar-a(-ba)*, *at-tar-i*, *ar-ō* and *ar-e* — thereby dividing the string into two words, the following alternations in (21) can be observed:

- | | |
|---|--|
| (21) <i>taka-k+at-ta</i> $\approx$ <i>taka-ku=mo#at-ta</i> | 'It was high/tall' $\approx$ '... also high/tall' |
| <i>taka-k+at-tar-ō</i> $\approx$ <i>taka-ku=mo#at-tar-ō</i> | 'It would be high/tall' $\approx$ '... also high/tall' |
| <i>taka-k+at-tar-a(-ba)</i> $\approx$ <i>taka-ku=mo#at-tar-a(-ba)</i> | 'If it is/were high/tall' $\approx$ '... also high/tall' |
| <i>taka-k+at-tar-i</i> $\approx$ <i>taka-ku=mo#at-tar-i</i> | 'Sometimes it is high/tall' $\approx$ '... also high/tall' |
| <i>taka-k+ar-ō</i> $\approx$ <i>taka-ku=mo#ar-ō</i> | 'It will be high/tall' $\approx$ '... also high/tall' |
| <i>taka-k+ar-e</i> $\approx$ <i>taka-ku=mo#ar-e</i> | 'Be high/tall' $\approx$ '... also high/tall' |

On this basis, it can be seen that the VA thematic suffix *-ku* undergoes vowel deletion and surfaces as *-k* when it precedes the vowel-initial stem of the light verb *ar-u*, namely *ar- ≈ at-*. This vowel deletion may be taken to indicate that **the VA stem in *-ku* and the light verb stem *ar-*** — each of which can function as an autonomous word — have come to **form a single compound stem**.

The reason why the inflected forms in (21) contain *ar-* or *at-* — stem allomorphs of the light verb *ar-u* — is that the endings *-ta*, *-tar-ō*, *-tar-a(ba)*, *-tar-i*, *-ō* and *-e*, which mark tense, modality or both, can combine only with purely verbal stems. In other words, *takaku* is a non-verbal stem and thus cannot combine directly with these endings. Consequently, the VA stem *taka-ku* combines with the stem *ar-* of the light verb *ar-u*, thereby forming a **compound V stem *taka-k+ar-*** (← *taka-ku* + *ar-*) at the sentential level, which as a whole comes to **possess verbal status and be able to combine with the verbal endings *-ta*, *-tar-ō*, *-tar-a(-ba)*, *-tar-i*, *-ō* and *-e***. This is made possible by the fact that Japanese is a head-final language, whereby the light verb stem *ar-* determines the syntactic category of the entire compound stem *taka-k+ar-*. Under this interpretation, the structure of (14) may be more accurately represented as in (22), where '-', '()' and '+' with no spaces on either side indicate suffixation, deletion and compounding, respectively, as in (1):

(22)

Word			
Stem: SENTENTIAL			Ending
VA stem: LEXICAL		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-ku</i>	—	—
<i>taka-</i>	<i>-ku</i>	—	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-a(-ba)</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-e</i>

Besides the allomorphy *-ku ≈ -k*, which is conditioned both morphosyntactically and phonologically, the VA thematic suffix exhibits another type of allomorphy, *-ku ≈ -i ≈ -(k)i ≈ -ke*, depending on the syntactic environment in which it occurs, as shown in (23)–(28), where LOC and GEN stand for *location* and *genitive*, respectively:

- (23) *kanoz_{yo}=wa se=ga *taka*
 she(POSS)=TOP stature(THEM)=NOM be_high
 'She is tall.' (ungrammatical)

- (24) *kanozyo=wa se=ga taka-i*
 she(POSS)=TOP stature(THEM)=NOM be_high-NPST
 'She is tall.' (grammatical)
- (25) *asoko=ni se=no *taka otoko=ga ir-u*
 there(LOC)=POST stature(THEM)=GEN be_high man(POSS/THEM)=NOM exist-NPST
 'There is a tall man over there.' (ungrammatical)
- (26) *asoko=ni se=no taka-i otoko=ga ir-u*
 there(LOC)=POST stature(THEM)=GEN be_high-NPST man(POSS/THEM)=NOM exist-NPST
 'There is a tall man over there.' (grammatical)
- (27) *watasi=wa nedan=ga *taka-reba kawa- -nai*
 he(AGT)=TOP price(THEM)=NOM be_high-COND buy NEG-NPST
 'I, if the price is high, do not buy (it).' (ungrammatical)
- (28) *watasi=wa nedan=ga taka-ke-reba kawa- -nai*
 he(AGT)=TOP price(THEM)=NOM be_high-COND buy NEG-NPST
 'I, if the price is high, do not buy (it).' (grammatical)

Among these, *taka-ke* in (28), which bears the allomorph *-ke*, occurs in Contemporary Japanese exclusively before the suffix *-reba*, which marks a sequential conditional: hence, *taka-ke-reba* is referred to as *kateikei* — namely, the conditional form. By contrast, the syntactic conditions that determine the occurrence of *taka-i* in (24) and *taka-(k)i* in (26), which bear the allomorphs *-i* and *-(k)i*, respectively, will be examined in the following section.

4. Syntactic conditions governing the allomorphy: *-ku* $\approx$ *-i* $\approx$ *-(k)i* $\approx$ *-ke*

As observed in (4), the conjunctive/adverbial form *taka-ku*, in which the VA thematic suffix appears in the form *-ku* and no inflexional ending follows, is **atemporal**. The fact that the form *taka-ku* lacks temporal reference is consistent with the interpretation that it does not bear any inflexional suffix marking tense, modality or both. By contrast, the conclusive form *taka-i* and the adnominal form *taka-(k)i* — the so-called *rentaikei* — appear to lack any inflexional ending just as the conjunctive/adverbial form *taka-ku* does, yet unlike *taka-ku*, they bear a **non-past value**. The question, then, is how this non-past value is formally encoded.

Traditionally, the final *-i* in *taka-i* and *taka-(k)i* has often been interpreted as the word constituent that bears the temporal value of non-past. This interpretation is difficult to justify, however, both historically and in light of the classical yet fundamental thesis that **linguistic value arises from opposition**, as Saussure (1916: 187; 196) puts it: "Dans l'intérieur d'une même langue, tous les mots qui expriment des idées voisines se limitent réciproquement ... n'ont de valeur propre que par leur opposition"; "deux signes comportant chacun un signifié et un signifiant ... sont

seulement distincts. Entre eux il n'y a qu'opposition. Tout le mécanisme du langage ... repose sur des oppositions de ce genre et sur les différences phoniques et conceptuelles qu'elles impliquent".

For the *-i* in *taka-i* to acquire the [–PAST] value, it must be paradigmatically opposed to the past suffix *-ta*. This would require the existence of a form **taka-ta*, in the same way that the suffix *-ru* in *tabe-ru* acquires the [–PAST] value only by being opposed to the suffix *-ta* in *tabe-ta*, but in fact no such form as **taka-ta* exists. Conversely, for the *-ta* in the attested form *takakat-ta* to acquire the [+PAST] value in opposition to *-i*, the hypothetical form **takakat-i* would be required. To what, then, is the *-i* in *taka-i* actually opposed in order to acquire the [–PAST] value?

Historically, verbal adjectives originally lacked a non-past vs. past tense opposition and, accordingly, they contained no formal element that encoded tense. In eighth-century Old Japanese, forms such as *taka-si*, *taka-ki* and *taka-ku* consisted, from the standpoint of the present analysis, solely of the stem, lacking any inflexional ending. However, probably due to the increasing need to encode the past value more explicitly, the past tense suffix *-ki* — which was solely compatible with purely verbal stems — and the stem allomorph *ari-* with epenthetic *-i* of the light verb *ari* 'be' were employed to constitute the past tense form *taka-k+ari-ki*.

As Okimori (2017: 84, translated from Japanese) states, "The secondary conjugation known as *kari*-conjugation was formed by the conjunctive/adverbial form being followed by the *r*-irregular verb *ari*, through the development *ku-ari* > *kari*. This made it possible for the form to attach to a wide range of particles and auxiliary verbs". Frellesvig (2010: 90) likewise notes: "In addition to its basic inflected forms, the adjectival copula formed extended analytic forms by the infinitive *-ku* followed by *ar-* ...: *-ku ar-*. ... This allowed the formation of ... combination with auxiliaries which did not combine directly with the adjectival copula.... The analytic formations provided full morphophonological versatility for the adjectival copula. The analytic forms sometimes fused phonologically, *-ku ar-* > *-kar-*, which in EMJ gave rise to a secondary conjugation of the adjectival copula".

This suggests that the conclusive form *taka-si* and the adnominal form *taka-ki*, which originally lacked any overt tense marker, came to **bear a non-past value** through their formal opposition to the newly constructed past form *taka-k+ari-ki* — that is, **by lacking the overt tense suffix *-ki***. In other words, from the perspective of formal opposition, those two forms came to be analysed as bearing a **phonologically null inflexional ending *-Ø* specified as [–PAST]** as in [[[*taka*]*si*]*Ø*] and [[[*taka*]*ki*]*Ø*]. This null ending came to stand in opposition to the past-tense inflexional suffix *-ki* specified as [+PAST] as in [[[*taka*]*k+ari*]*ki*], which is consistent with the Saussurean view that "La langue est un système de pures valeurs que rien ne détermine en dehors de l'état momentané de ses termes" (Saussure 1916: 132). This is parallel to the case of English *cat-Ø*, where the lack

of an overt plural marker in opposition to *cat-s* functions to mark the singular. Based on the above, the morphosyntactic structure of the Early Middle Japanese (EMJ) forms *taka-si*, *taka-ki*, *taka-ku* and *taka-k+ari-ki* may be analysed as in (29):

(29)

Word			
Stem: SENTENTIAL			Ending
VA stem: LEXICAL		Light verb stem	Inflexional suffix
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-si</i>	—	-Ø [-PAST]
<i>taka-</i>	<i>-ki</i>	—	-Ø [-PAST]
<i>taka-</i>	<i>-ku</i>	—	— [Ø PAST]
<i>taka-</i>	<i>-k(u)</i>	+ari-	-ki [+PAST]

In Contemporary Japanese as well, it may be assumed that the partial system in (29) remains unchanged, except that the past-tense suffix *-ki* has simply been replaced by *-ta*, as shown in (30):

(30)

Word			
Stem: SENTENTIAL			Ending
VA stem: LEXICAL		Light verb stem	Inflexional suffix
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i>	—	-Ø [-PAST]
<i>taka-</i>	<i>-(k)i</i>	—	-Ø [-PAST]
<i>taka-</i>	<i>-ku</i>	—	— [Ø PAST]
<i>taka-</i>	<i>-k(u)</i>	+at-	-ta [+PAST]

The remaining issues to be discussed are the relationship between the conclusive form *taka-i* and the adnominal form *taka-(k)i*, and the conditions that determine their distribution in relation to the conditional form *taka-ke* and the conjunctive/adverbial form *taka-ku*. The first two forms, *taka-i* and *taka-(k)i*, share the VA radical *taka-* and a zero tense suffix *-Ø* specified as [-PAST], but differ in the shape of the VA thematic suffix: *-i* in the conclusive form and *-(k)i* in the adnominal form. Correspondingly, they occur in distinct syntactic environments, as in (31) and (32):

- (31) *utukusi-(k)i* hito
 be_beautiful-VA.TH-NPST person
 'A beautiful person'

[_{NP} [_{CP} [_{TP} [_{VAP} *op_i* *utukusi-(k)i*] -Ø] -Ø] *pito_i*] → [_{NP} [_{CP} [_{TP} [_{VAP} *op_i* *utukusi-(k)i*] -Ø] -Ø] *hito_i*]³

³ There is also the possibility of [_{NP} [_{CP} [_{TP} [_{VAP} *op_i* *utukusi-(k)i*] -Ø] -Ø] *hito_i*] → [_{NP} [_{CP} *op_i* [_{TP} [_{VAP} *t_i* *utukusi-(k)i*] -Ø] -Ø] *hito_i*]. However, this paper employs a simplified model focusing on the allomorphy of the VA thematic suffix *-ku* ≈ *-i* ≈ *-(k)i* ≈ *-ke* and on its syntactic conditions, leaving detailed syntactic verification to future work.

- (32) *kanozyo=wa utukusi-i*
 she(THEM)=TOP be-beautiful-VA.TH-NPST
 'She is beautiful.'

[_{TopP} [_{TP} [_{VAP} *kanozyo utukusi-i*] -Ø] -Ø] → [_{TopP} *kanozyo_i=wa* [_{TP} [_{VAP} *t_i utukusi-i*] -Ø] -Ø]

As shown in (31), when a VAP functions as the complement of TP headed by T -Ø [-PAST] and this TP is embedded within an NP, the VA thematic suffix appears in the form *-(k)i*. By contrast, when a VAP functions as the complement of TP headed likewise by T -Ø [-PAST] but the TP is not embedded within an NP, as in (32), the VA thematic suffix appears in the form *-i*. As for the remaining forms *-ke* and *-ku*, *-ke* in Contemporary Japanese is exclusively distributed in the environment immediately before the TP-head conditional suffix *-reba* as in (33):

- (33) *kanozyo=ga utukusi-ke-reba*
 she(THEM)=NOM be_beautiful-VA.TH-COND
 'If she is beautiful'

[_{TP} [_{AP} *kanozyo utukusi-ke*] -reba] → [_{TP} *kanozyo_i=ga* [_{AP} *t_i utukusi-ke*] -reba]

while *-ku* is the form taken by the VA thematic suffix in all other syntactic environments — namely **the *elsewhere* form** — when the conditions for the more specific forms *-i*, *-(k)i* and *-ke* are not met. Thus, it seems **more appropriate to refer to VAs as *ku*-adjectives** rather than *i*-adjectives, after their **default form in *-ku*** — diachronically the earliest form in which VAs were used as syntactic constituents at the sentential level, as Okimori (2017: 85–86, translated from Japanese) states: "Adjectives originally had as their basic form only the conjunctive form used adverbially. The conjunctive inflexional ending *-ku* was ... an adverbial ending and the conjunctive form was originally an adverb. ... That adverb came to function like a predicate and came to have the use of the suspensive form (*tyūsikei*).\" Finally, on the view adopted in this paper that the formation of VA stems is effected in the lexicon, the syntactic derivations in (31)–(33) and the allomorphy of the VA thematic suffix *-ku* can, more precisely, be represented as in (34):

- (34) [_{NP} [_{CP} [_{TP} [_{VAP} *op_i utukusi-ku*] -Ø] -Ø] *pito_i*] → [_{NP} [_{CP} [_{TP} [_{VAP} *op_i utukusi-(k)i*] -Ø] -Ø] *hito_i*]
 [_{TopP} [_{TP} [_{VAP} *kanozyo utukusi-ku*] -Ø] -Ø] → [_{TopP} *kanozyo_i=wa* [_{TP} [_{VAP} *t_i utukusi-i*] -Ø] -Ø]
 [_{TP} [_{AP} *kanozyo utukusi-ku*] -reba] → [_{TP} *kanozyo_i=ga* [_{AP} *t_i utukusi-ke*] -reba]

5. An alternative interpretation of the allomorphy: *-ku* ≈ *-i* ≈ *-(k)i* ≈ *-ke*

Building on the structuralist–functionalist notion of *amalgam*, another analysis for the allomorphy *-ku* ≈ *-i* ≈ *-(k)i* ≈ *-ke* becomes possible. Regarding the notion of amalgam — *amalgame* in French

— Martinet (1991: 102) writes: "deux signifiés qui coexistent dans un énoncé enchevêtrent leurs signifiants de telle façon qu'on ne saurait analyser le résultat en segments successifs".

As is clear from the description in the final paragraph of Section 4, the allomorph *-ku* of the VA thematic suffix, unlike *-i*, *-(k)i* and *-ke*, which occur under specific morphosyntactic conditions as in (31)–(33), can be regarded as the default form — namely, the form that surfaces when no specific morphosyntactic condition is met, or the *elsewhere* form. Furthermore, whereas *-ku* can occur on its own without any following morpheme, *-i* and *-(k)i* always appear with a specific morpheme following them — namely, $-\emptyset$ [–PAST]. From this, the amalgams in (35) become conceivable:

(35) *-ku* + [–PAST] → *-i* or *-(k)i*, depending on morphosyntactic conditions.

Under this analysis, ***-i* and *-(k)i* are, synchronically, not allomorphs exclusively of the VA thematic suffix**, but **simultaneously encode the value [–PAST]**, thereby *partially* overlapping with the traditional analysis. It should nevertheless be noted that, as indicated in Section 4, ***-i* and *-(k)i* cannot be interpreted as allomorphs exclusively of the tense suffix [–PAST] either**, in light of the Saussurean thesis that linguistic value arises from opposition.

6. Structural similarities with inflected verbal forms

This section examines the inflected forms of the vowel-stem verb *nobiru* 'stretch/extend' (non-causative) and its EMJ form *nobu*, and demonstrates the significant structural parallels between these forms and the inflected forms of the verbal adjective *takai* 'be high/tall' and its EMJ form *takasi*.

The one-argument verb *nobiru* is a unaccusative one, with the stem *nobi(-)*. By contrast, the morphosyntactically and semantically corresponding two-argument and accusative verb *noberu* 'stretch/extend' (causative) has the stem *nobe(-)*. The shared element *nob-* in both the non-causative verb stem *nobi(-)* and the causative verb stem *nobe(-)* is present in all the inflected forms of these two verbs. Moreover, since the meaning of 'stretching/extension' as a state change is common to all their inflected forms, it can be concluded that *nob-* signifies the state change of 'stretching/extension' itself. In modern Japanese, although the causative verb corresponding to *nobiru* is more commonly *nobasu* — which can be analyzed as underlyingly *nob-sas-ru* with the causative morpheme *-sas-*, surfacing as *nob-as-u* — the complex stem *nob-as-* itself also contains the element *nob-* in its inflected forms, signifying the same state change.

Thus, the stem *nobi(-)* contains a morpheme boundary within it, as in *[[nob]i]*. Since neither *nob-* nor *-i* can be further segmented into smaller sequential units, *nob-* is one of the two immediate constituents of the verbal stem *nob-i(-)* and, encoding the core lexical meaning of the verb, can be considered, alongside the VA radical discussed in Section 3, as the radical of the verb — the **V radical**. On the other hand, the *-i* in *nob-i(-)* combines with the V radical *nob-* to form the stem of the verb — the **V stem** — and is assumed to function to license the V radical as the category V at the sentential level. Thus, in parallel with the VA thematic suffix *-ku*, this *-i* can be considered the thematic suffix of the verb — the **V thematic suffix**.

This suffix *-i* in *nob-i(-)* is also assumed to confer causativity on the V radical *nob-* through its opposition to *-e* in the causative verb stem *nob-e(-)*, which, like *-i*, is a V thematic suffix and is assumed to confer causativity on the V radical *nob-*. Thus, the valency of the V stem *nob-i(-)* and that of the V stem *nob-e(-)* are determined by the argument structure inherent in the V radical *nob-* itself and by whether the selected V thematic suffix adds a causer to that structure. Since the argument structure of *nob-* is [theme], *nob-i(-)* retains [theme], whereas *nob-e(-)* has [causer, theme].

Based on the above analysis, the morphosyntactic structure of the following inflected forms of the VA *takai* — *taka-i-Ø*, *taka-(k)i-Ø*, *taka-ku*, *taka-ke-reba* and *taka-k+at-ta* — and their corresponding forms of the V *nobiru* — *nob-i-ru*, *nob-i-ru*, *nob-i*, *nob-i-reba* and *nob-i-ta* — is as shown in (36), where '##' stands for a main-clause-final boundary:

(36)

Word			
Stem: SENTENTIAL			Ending
Stem: LEXICAL		Light verb stem	Inflexional suffix
Radical	Thematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-Ø##</i>
<i>nob-</i>	<i>-i</i>	—	<i>-ru##</i>
<i>taka-</i>	<i>-(k)i</i>	—	<i>-Ø#NP</i>
<i>nob-</i>	<i>-i</i>	—	<i>-ru#NP</i>
<i>taka-</i>	<i>-ku</i>	—	—
<i>nob-</i>	<i>-i</i>	—	—
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i> 'if'
<i>nob-</i>	<i>-i</i>	—	<i>-reba</i> 'if'
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i>
<i>nob-</i>	<i>-i</i>	—	<i>-ta</i>

Apart from the difference whereby the tense [–PAST] is encoded by *-Ø* in the VA *taka-i-Ø##* but by *-ru* in the V *nob-i-ru##*, and apart from the possible insertion, in VA forms, of the stem *at-* of the light verb *ar-u* at the sentential level, the basic structure is the same.

The inflected forms of the corresponding EMJ forms *takasi* and *nobu* exhibit even greater structural similarity, as indicated in (37), where the realis forms — so-called *izenkei* — *taka-ke-re*, *nob-u-re*, *taka-ke-re-do(mo)* and *nob-u-re-do(mo)* are added:

(37)

Word			
Stem: SENTENTIAL		Ending	
Stem: LEXICAL		Light verb stem	Inflexional suffix
Radical	Thematic suffix		
<i>taka-</i>	<i>-si</i>	—	<i>-Ø##</i>
<i>nob-</i>	<i>-u</i>	—	<i>-Ø##</i>
<i>taka-</i>	<i>-ki</i>	—	<i>-Ø#NP</i>
<i>nob-</i>	<i>-u</i>	—	<i>-ru#NP</i>
<i>taka-</i>	<i>-ku</i>	—	—
<i>nob-</i>	<i>-i</i>	—	—
<i>taka-</i>	<i>-ke</i>	—	<i>-re</i>
<i>nob-</i>	<i>-u</i>	—	<i>-re</i>
<i>taka-</i>	<i>-ke</i>	—	<i>-re-ba</i> 'since/when'
<i>nob-</i>	<i>-u</i>	—	<i>-re-ba</i> 'since/when'
<i>taka-</i>	<i>-ke</i>	—	<i>-re-do(mo)</i> 'though'
<i>nob-</i>	<i>-u</i>	—	<i>-re-do(mo)</i> 'though'
<i>taka-</i>	<i>-k(u)</i>	<i>+ari-</i>	<i>-ki</i>
<i>nob-</i>	<i>-i</i>	—	<i>-ki</i>

In (37), what corresponds diachronically to the CJ conditional suffix *-reba* has to be analysed as a sequence of two morphemes *-re-ba*, since, in EMJ, *-re-ba* is morphosyntactically and semantically opposed, on the one hand, to *-Ø-ba* and, on the other, to *-re-do(mo)*, as in *m-i-re-ba* 'since/when one sees' in opposition to *m-i-Ø-ba* 'if one sees' and *m-i-re-do(mo)* 'though one sees', from the verb *m-i-ru* 'look at; see'. It follows, therefore, that *-re* and *-ba* in *-re-ba* are two distinct morphemes with specific semantic values of their own.

In Makino (2026), *-re* in *-re-ba* and *-re-do(mo)* is analysed as a contextual allomorph of the suffix *-ru*, selected in the context *_-ba*, *-do(mo)*, on the grounds that *-ru* and *-re* are complementarily distributed and bear the same modal function of presenting the eventuality expressed by the proposition as belonging to the actual world. Likewise, *-re* in bare realis forms in *-re* without the conditional suffix *-ba* nor the concessive suffix *-do(mo)*, employed in the *[[...=koso] + [... -re]]* type of the so-called *kakari-musubi*, as in *...=koso ... m-i-re* 'one sees ... precisely ...', is also analysed as the same contextual allomorph. In this construction, *-ru* is assumed to surface as *-re* in the context *_-Ø* [+PREM(ISE-SETTING)], as in *m-i-ru-Ø* → *m-i-re-Ø*, where *-Ø* sets the proposition marked by *-ru* as a premise for subsequent focalisation by the focus particle *=koso*. Furthermore, in EMJ bigrade (*nidan*) verbs such as *nob-u*, the suffix *-ru* is realised as *-Ø* in the

context *_##* — where *##* stands for a main-clause-final boundary — as demonstrated by the alternation *nob-u-ru#NP* $\approx$ *nob-u-Ø##*.

Thus, the suffix *-ru* surfaces as *-ru*, *-Ø* or *-re*, depending on the context. Under this analysis, the alternation of the V thematic suffix *-i* to *-u* in (37) occurs only immediately before the suffix *-ru*. Therefore, *-u* can be assumed to be a contextual allomorph of the V thematic suffix *-i*. On this basis, if the underlying forms (Ufs) of *nob-u-Ø##*, *nob-u-ru#NP*, *nob-u-re-Ø* [+PREM], *nob-u-re-ba* and *nob-u-re-do(mo)* are assumed to be *nob-i-ru##*, *nob-i-ru#NP*, *nob-i-ru-Ø*, *nob-i-ru-ba* and *nob-i-ru-do(mo)*, respectively, the surface forms (SFs) can be derived by the allomorphies: *-i* $\rightarrow$ *-u* / *_ru*; *-ru* $\rightarrow$ *-Ø* / *_##*; and *-ru* $\rightarrow$ *-re* / *_Ø* [+PREM], *-ba*, *-do(mo)*⁴. If, in parallel with the allomorphy *-ru* $\rightarrow$ *-Ø* / *_##* in bigrade verbs, it is assumed that the allomorphy *-ru* $\rightarrow$ *-Ø* / *_##*, and further *-ru* $\rightarrow$ *-Ø* / *_#NP*, also takes place in VAs, then the UFs of *taka-i-Ø##*, *taka-ki-Ø#NP*, *taka-ke-re-Ø* [+PREM], *taka-ke-re-ba* and *taka-ke-re-do(mo)* can be assumed to be *taka-ku-ru##*, *taka-ku-ru#NP*, *taka-ku-ru-Ø*, *taka-ku-ru-ba* and *taka-ku-ru-do(mo)*, respectively, as illustrated in (38), where Rad., Them.suff. and Infl.suff. stand for *radical*, *thematic suffix* and *inflexional suffix*, respectively:

(38)

Underlying form					$\Rightarrow$	Surface form				
Stem		Ending				Stem		Ending		
Rad.	Them.suff.	Infl.suff.	Infl.suff.			Rad.	Them.suff.	Infl.suff.	Infl.suff.	
<i>taka-</i>	<i>-ku</i>			#		<i>taka-</i>	<i>-ku</i>			#
<i>nob-</i>	<i>-i</i>			#		<i>nob-</i>	<i>-i</i>			#
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>		##		<i>taka-</i>	<i>-si</i>	<i>-Ø</i>		##
<i>nob-</i>	<i>-i</i>	<i>-ru</i>		##		<i>nob-</i>	<i>-u</i>	<i>-Ø</i>		##
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>		#NP		<i>taka-</i>	<i>-ki</i>	<i>-Ø</i>		#NP
<i>nob-</i>	<i>-i</i>	<i>-ru</i>		#NP		<i>nob-</i>	<i>-u</i>	<i>-ru</i>		#NP
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>	<i>-Ø</i>	#		<i>taka-</i>	<i>-ke</i>	<i>-re</i>	<i>-Ø</i>	#
<i>nob-</i>	<i>-i</i>	<i>-ru</i>	<i>-Ø</i>	#		<i>nob-</i>	<i>-u</i>	<i>-re</i>	<i>-Ø</i>	#
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>	<i>-ba</i>	#		<i>taka-</i>	<i>-ke</i>	<i>-re</i>	<i>-ba</i>	#
<i>nob-</i>	<i>-i</i>	<i>-ru</i>	<i>-ba</i>	#		<i>nob-</i>	<i>-u</i>	<i>-re</i>	<i>-ba</i>	#
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>	<i>-do(mo)</i>	#		<i>taka-</i>	<i>-ke</i>	<i>-re</i>	<i>-do(mo)</i>	#
<i>nob-</i>	<i>-i</i>	<i>-ru</i>	<i>-do(mo)</i>	#		<i>nob-</i>	<i>-u</i>	<i>-re</i>	<i>-do(mo)</i>	#

The inflected forms of *takasi* and *nobu* in (38) exhibit a high degree of structural similarity. Just as the VA thematic suffix *-ku* shows the allomorphy *-ku* $\rightarrow$ *-si*, *-ki*, *-ke*, depending on the syntactic environment, the V thematic suffix *-i* also exhibits similar allomorphy, namely *-i* $\rightarrow$ *-u*. Both conjunctive forms *taka-ku* and *nob-i* lack any inflexional suffix. The suffix *-ru* is realised as phonologically null *-Ø* in the VA conclusive form *taka-si-Ø##*, as well as in the corresponding V

⁴ In Makino (2026), the allomorphy *-ru* $\rightarrow$ *-re* / *_Ø* [+PREM], *-ba*, *-do(mo)* is generalised as *-ru* $\rightarrow$ *-re* / PREMISE-SETTING ENVIRONMENT: *-Ø*, *-ba*, *-do(mo)*, since a proposition marked by one of these suffixes is incorporated as a premise into another semantic structure: a focus-background structure in the case of *-Ø* [+PREM], a conditional structure in the case of *-ba* and a concessive structure in the case of *-do(mo)*.

form *nob-u-Ø##*. This suffix is consistently realised as *-re* before the suffixes *-Ø* [+PREM], *-ba* and *-do(mo)* not only in V realis forms but also in VA realis forms. Furthermore, the VA realis forms such as *taka-ke-re*, *taka-ke-re-ba* and *taka-ke-re-do(mo)* are the only VA inflected forms in which **the suffix *-ru* is realised overtly**. The existence of such forms allows us to posit the underlying structure common to the conclusive, adnominal and realis forms as *taka-ku-ru-*. The only instance in which the parallelism breaks down is in the adnominal forms *taka-ki-Ø#NP* and *nob-u-ru#NP*, but there seems to be a reason for this. In the VA *takasi*, the opposition between the conclusive and adnominal forms was maintained by the alternation of the thematic-suffix allomorphs *-si* $\approx$ *-ki*, as in *taka-si-Ø##* and *taka-ki-Ø#NP*, whereas in the V *nobu*, the thematic-suffix allomorph was the same *-u* in both forms and the opposition is instead maintained by the alternation of the modal-suffix allomorphs *-Ø* $\approx$ *-ru*, as in *nob-u-Ø##* and *nob-u-ru#NP*.

7. Negative forms of verbal adjectives

Both verbs and verbal adjectives form their negative constructions by attaching the negator *nai* to their stems, as in *tabe-nai* 'not eat' from the V *tabe-ru* and *takaku#nai* 'not be high/tall' from the VA *takai*. This negator belongs to the VA category, as demonstrated by its inflexion — namely, *na-i* $\approx$ *na-ku* $\approx$ *na-ku-te* $\approx$ *na-ku-temo* $\approx$ *na-ku-tatte* $\approx$ *na-k+at-ta* $\approx$ *na-k+at-tar-ō* $\approx$ *na-k+at-tar-a(-ba)* $\approx$ *na-k+at-tar-i* $\approx$ *na-k+ar-ō* $\approx$ *na-ke-reba*, as in *tabe-na-i* $\approx$ *tabe-na-ku* $\approx$ *tabe-na-ku-te* $\approx$ *tabe-na-ku-temo* $\approx$ *tabe-na-ku-tatte* $\approx$ *tabe-na-k+at-ta* $\approx$ *tabe-na-k+at-tar-ō* $\approx$ *tabe-na-k+at-tar-a(-ba)* $\approx$ *tabe-na-k+at-tar-i* $\approx$ *tabe-na-k+ar-ō* $\approx$ *tabe-na-ke-reba* and *takaku#na-i* $\approx$ *takaku#na-ku* $\approx$ *takaku#na-ku-te* $\approx$ *takaku#na-ku-temo* $\approx$ *takaku#na-ku-tatte* $\approx$ *takaku#na-k+at-ta* $\approx$ *takaku#na-k+at-tar-ō* $\approx$ *takaku#na-k+at-tar-a(-ba)* $\approx$ *takaku#na-k+at-tar-i* $\approx$ *takaku#na-k+ar-ō* $\approx$ *takaku#na-ke-reba*. This inflexional pattern is identical to that of the verbal adjective *takai* — namely, *taka-i* $\approx$ *taka-ku* $\approx$ *taka-ku-te* $\approx$ *taka-ku-temo* $\approx$ *taka-ku-tatte* $\approx$ *taka-k+at-ta* $\approx$ *taka-k+at-tar-ō* $\approx$ *taka-k+at-tar-a(-ba)* $\approx$ *taka-k+at-tar-i* $\approx$ *taka-k+ar-ō* $\approx$ *taka-ke-reba*.

Although the above negative forms *tabe-nai* and *takaku#nai* may at first sight appear to exhibit the same morphosyntactic structure, in that they both consist of "stem + negator *nai*", their structures in fact differ: **the negator *nai* in *tabe-nai* functions as a sequence of suffixes *-na-i***, whereas ***nai* in *takaku#nai* constitutes the autonomous word *#na-i***, as illustrated in (39):

(39) <i>tabe-nai</i>	$\rightarrow$ <i>tabe=wa#<u>si-nai</u></i>	vs. * <i>tabe=wa#<u>nai</u></i>	'eat, it does not'
<i>tabe-nai</i>	$\rightarrow$ <i>tabe=mo#<u>si-nai</u></i>	vs. * <i>tabe=mo#<u>nai</u></i>	'it does not even eat'
<i>takaku#nai</i>	$\rightarrow$ <i>takaku=wa#<u>nai</u></i>	vs. * <i>takaku=wa#<u>ara-nai</u></i>	'high/tall, it is not'

takaku#nai → *takaku=mo#nai* vs. **takaku=mo#ara-nai* 'it is not high/tall either'
 cf. *takaku-te* → *takaku=mo#at-te* vs. **takaku=mo#te* in (15)

In both *tabe-nai* and *takaku#nai*, the enclitic particle =*mo* or =*wa* can be syntactically inserted immediately after the V stem *tabe-* and the VA stem *takaku-*, respectively, as in the examples in (15) and (16) above. In the former case, the stem allomorph *si-* of the light verb *su-ru* must be inserted before the negator *nai*, as in *tabe=wa#si-nai* and *tabe=mo#si-nai*. The reason is as follows: since *nai* functions as a sequence of suffixes *-na-i* with respect to the stem *tabe-* in *tabe-nai*, once the particle =*mo* or =*wa* is inserted between the stem *tabe-* and the suffixal sequence *-na-i*, the latter is left stemless. Consequently, an alternative formal stem must be supplied — namely, *si-* from *su-ru* — since =*mo* and =*wa* cannot function as verbal stems for *-na-i*.

In the latter case, by contrast, the same particle insertion does not entail the insertion of the stem allomorph *ara-* from the light verb *ar-u* before *nai*, as in **takaku=wa#ara-nai* and **takaku=mo#ara-nai* — the allomorph expected in this morphosyntactic context for consonant-stem verbs such as *ar-u*, just as in *noma-nai* from the consonant-stem verb *nom-u* 'drink'. This indicates that, in *takaku#nai*, the string *na-i* functions as an autonomous word in relation to the other autonomous word *takaku* and therefore requires no verbal stem before it. The negator *nai* in *takaku#nai* is morphologically distinct, for example, from the suffix *-te* in *takaku-te*, which requires the stem allomorph *at-* from *ar-u* in the particle-inserted form *takaku=mo#at-te*, as already shown in (15).

The above reasoning is based on the insightful observation in Miyaoka (2002: 85, translated from Japanese and noted by the author): "whereas in *kaka-nai* 'it does not write' the suffixality of *-nai* is evident, since *si* (< *suru* 'do') must be inserted as in *kaki=mo=si-nai* 'it does not even write', in *takaku=nai* the possibility of inserting the clitic =*mo* directly shows the cliticity of =*nai*."

The above reasoning is based on the insightful observation in Miyaoka (2002: 85, translated from Japanese): "whereas in *kaka-nai* 'it does not write' the suffixality of *-nai* is evident, since *si* (< *suru*) must be inserted as in *kaki=mo=si-nai* 'it does not even write', in *takaku=nai* the possibility of directly inserting the clitic =*mo* shows the cliticity of =*nai*."

The question arises, then, why the negator *nai* functions as a string of suffixes *-na-i* in *tabe-nai* but as an autonomous word #*na-i* in *takaku#nai*. In other words, why is there a lack of parallelism in morphosyntactic structure between the negative forms of verbs and those of verbal adjectives? This seems to be related to the fact that, in Contemporary Japanese, the negative form of the verb *ar-u* is not the theoretically expected **ara-nai* — composed of the stem allomorph *ara-* from *ar-u* and the suffixes *-na-i*, as in *noma-nai* from *nom-u* and *tabe-nai* from *tabe-ru* — but rather simply *nai*, which is categorially not a verb but a verbal adjective, as *takai* is. In other words, **the**

VA *nai* is a kind of suppletive form corresponding to the hypothetical negative form **ara-nai* within the inflexional or derivational paradigm of the verb *ar-u*.

Building on this suppletion, let us hypothesise that, in the negation of verbal adjectives such as *takai*, it is likewise the suffixal sequence *-na-i* used for verbs — rather than the autonomous word *#na-i* — that is employed underlyingly, just as in the case of verbal negation. The derivation from *takaku* + *-nai* to the attested SF *takaku#nai* is then assumed to proceed as shown in (40), where '→' indicates a failed derivation:

- (40) *takai* + *-nai* → **takaku#ara-nai* → *takaku#nai*: suppletion
 → **takak+ara-nai*: no suppletion + vowel deletion

Since the verbal-adjective stem *takaku-* is not compatible with the negative suffix *-na*, which is verbal and subcategorises for purely verbal stems, the stem allomorph *ara-* from the light verb *ar-u* is inserted immediately after *takaku-*, thereby constituting, together with it, a composite verbal stem *takaku#ara-* at the level of syntactic derivation. This composite stem as a whole acquires verbal status, due to the head-finality of this language, and can combine directly with the negative suffix *-na*, yielding the hypothetical intermediate string **takaku#ara-nai* through the addition of the VA thematic suffix *-i* following *-na*. The hypothetical verbal negative form **ara-nai* in **takaku#ara-nai* is, however, replaced by the VA *nai*, yielding *takaku#nai*, before the vowel deletion that would yield **takak+ara-nai* can apply, owing to the obligatory suppletion described above within the inflexional or derivational paradigm of *ar-u*. At the same time, since the word boundary # preceding the replaced sequence remains intact, *takaku* and *nai* are realised as two autonomous words.

In Old Japanese, on the other hand, the negative form of the light verb *ari-Ø*, which corresponds to Modern Japanese *ar-u*, was not the VA *nasi*, which corresponds to Modern Japanese *nai*, but rather *ara-zu* with the negative suffix *-zu*, which corresponds to Modern Japanese *-nai* — just as the negative forms of *tabu-Ø*, corresponding to Modern Japanese *tabe-ru*, and *nom-u* were *tabe-zu* and *noma-zu*, respectively. Consequently, in the derivation of the negative form of the VA *takasi*, which corresponds to Modern Japanese *takai*, the negative form *ara-zu* was not replaced by *nasi*, as shown in (41):

- (41) *takasi* + *-zu* → *takaku#ara-zu* → *takak+ara-zu*: no suppletion + vowel deletion
 → **takaku#nasi*: suppletion

The EMJ form *takaku#ara-zu* in (41) — from which the more synthetic form *takak+ara-zu* originated — is the structural equivalent of the modern hypothetical **takaku#ara-nai* in (40),

whereas the EMJ hypothetical **takaku#nasi* in (41) is the structural equivalent of the modern attested *takaku#nai* in (40). Analysed in this way, the difference between Modern and Old Japanese in the formation of the negative forms of verbal adjectives lies in the modern replacement of the negative form **ara-nai* of the light verb *ar-u* by *nai* through suppletion, and the derivation schematised in (40) seems entirely plausible.

Thus, the verbal-adjective negative form *takaku#nai-Ø*, with the ending *-Ø* specified as [–PAST], is assumed to be derived from the intermediate form in (42):

(42) [[[*takaku#ara*]*nai*]*Ø*]

which exhibits the same structure as the verbal negative form in (43):

(43) [[[*tabe*]*nai*]*Ø*]

in that both strings are composed of [[[**verbal stem**]**suffixal negator**]**ending**], apart from the difference in the degree of complexity between the composite stem **takaku#ara-* and the simplex *tabe-*.

Based on the above discussion, the *surface* negative forms of the VA *takai* can be analysed as in (44), where R, TS and S represent the radical, the thematic suffix and the stem, respectively:

(44)

Word		Word			
VA S: SENTENTIAL		Negator S: SENTENTIAL			Ending
VA S: LEXICAL		Negator S: LEXICAL		Light verb stem	Inflexional suffix(es)
VA R	VA TS	Negator R	Negator TS		
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-i</i>	—	<i>-Ø</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-(k)i</i>	—	<i>-Ø</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i> [+PAST]
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-a(-ba)</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-ku</i>	<i>na-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-e</i>

8. Syntactic and logical-propositional differences between *ku*-adjectives and *ni*-adjectives

The Japanese language has another type of adjective — **nominal adjectives (NAs)**, also referred to as ***na*-adjectives** due to their adnominal forms ending in *-na* (the so-called *rentaikei* in Japanese linguistics), as in *sizuka-na heya* 'a room that is quiet'. According to Makino (2025¹), the core of NAs lies in the **NA stem in *-ni***, as in *sizuka-ni* 'be quiet/silent' and *suki-ni* 'like'. The element *-ni* is the NA thematic suffix, which functions as an **indispensable morphological and semantic constituent in the formation of the NA stem**. Thus, it seems **more appropriate to call NAs *ni*-adjectives** rather than *na*-adjectives, in accordance with this defining constituent *-ni*. NA stems are lexical predicates that require arguments — the argument structure [theme] inherent in *sizuka-ni* and [experiencer, theme] in *suki-ni* — just as VA stems do, as in the one-argument predicate *tanosi-ku* 'be enjoyable' with [theme] and the two-argument predicate *hosi-ku* 'want' with [experiencer, theme]. However, outside small clauses functioning as complements of cognition or perception verbs, as in (45) and (46), where EXP stands for *experiencer*:

- (45) *watasi=wa* [*sore=o* ***yukai-ni*** *omot-ta* / *kanzi-ta*
I(EXP)=TOP [it(THEM)=ACC be-pleasant] think-PST / feel-PST
'I found / felt [it pleasant].'
- (46) *watasi=wa* [*sore=o* ***tanosi-ku*** *omot-ta* / *kanzi-ta*
I(EXP)=TOP [it(THEM)=ACC be-enjoyable] think-PST / feel-PST
'I found / felt [it enjoyable].'

such NA stems cannot function as predicates in their bare *-ni* form, unlike VA stems, as shown in (47)–(50), where VA.TH and NA.TH stand for *VA thematic suffix* and *NA thematic suffix*, respectively:

- (47) *sore=wa* ***tanosi-ku*** (*sosite*) ***yukai-de*** *ar-u*
it(THEM)-TOP be-enjoyable-VA.TH (CONJ) be-pleasant-NA.TH-CONJ exist-NPST
'It, being enjoyable, (and) being pleasant, exists.'
- (48) *sore=wa* ***tanosi-ku*** (*sosite*) ****yukai-ni***
it(THEM)-TOP be-enjoyable-VA.TH (CONJ) be-pleasant-NA.TH
'It, being enjoyable, (and) being pleasant.'
- (49) *sore=wa* ***yukai-de*** *ari* (*sosite*) ***tanosi-i***
it(THEM)-TOP be-pleasant-NA.TH-CONJ exist (CONJ) be-enjoyable-VA.TH-NPST
'It, being pleasant, exists (and) is enjoyable.'
- (50) *sore=wa* ****yukai-ni*** (*sosite*) ***tanosi-i***
it(THEM)-TOP be-pleasant-NA.TH (CONJ) be-enjoyable-VA.TH-NPST
'It, being pleasant, (and) is enjoyable.'

Consequently, an NA stem forms a nominal-adjective phrase containing PRO, as in [*PRO yukai-ni*], which serves as the complement of the adjunctive-conjunction suffix *-te*, yielding a ConjP [[*PRO yukai-ni-te*]]. This ConjP is then adjoined to the VP headed by the existential-verb stem *ar-* within a one-argument obligatory construction, as in [*sore_i=wa t_i ar-ru*] 'it exists'. Only in this configuration can the NA stem function as one of the two predicates of an independent sentence, as in [*sore_i=wa [PRO_i yukai-ni-te] t_i ar-ru*], which surfaces as [*sore_i=wa [PRO_i yukai-de] t_i ar-u*] 'it is pleasant' — more literally, 'it, being pleasant, exists'.

This construction is schematically representable as $\text{EXIST}(\text{sore}) \wedge \text{PLEASANT}(\text{sore})$, and its default negation pattern is scope-limited, namely $\text{EXIST}(\text{sore}) \wedge \neg \text{PLEASANT}(\text{sore})$, with the scope of negation marked by the particle *=wa*, as indicated in (51) and (52):

- (51) *Sore=wa yukai-de ar-u*: $\text{EXIST}(\text{sore}) \wedge \text{PLEASANT}(\text{sore})$
 (52) *Sore=wa yukai-de=**wa** (*ara-)**nai***: $\text{EXIST}(\text{sore}) \wedge \neg \text{PLEASANT}(\text{sore})$

(Note that in (52) and (53) below, the stem of the existential verb *aru* is represented as *(*ara-)*, since the negative form of this verb is, as stated in Section 9, obligatorily the suppletive adjective *nai*.) The negative *-nai* in *(*ara-)**nai*** therefore does not negate the entire construction *sore=wa yukai-de ar-u*, but instead targets only the adjunct *yukai-de*. Semantically, the sentence continues to introduce the existence of *sore*, while negating only the claim that *sore* is *yukai-ni*. This follows from the adjunctive nature of *yukai-de*, which logically constitutes an additional proposition. The total negation in (53) is perceived as pragmatically infelicitous:

- (53) *Sore=wa yukai-de (*ara-)**nai***: $\neg(\text{EXIST}(\text{sore}) \wedge \text{PLEASANT}(\text{sore}))$

By contrast, the default negation pattern of the VA construction in (54) is the total negation in (55), not the scope-limited negation in (56):

- (54) *Sore=wa tanosi-i*: $\text{ENJOYABLE}(\text{sore})$
 (55) *Sore=wa tanosi-ku **nai***: $\neg \text{ENJOYABLE}(\text{sore})$
 (56) *Sore=wa tanosi-ku=**wa** **nai***: $\neg \text{ENJOYABLE}(\text{sore}) \wedge \text{SOME-OTHER-STATE}(\text{sore})$

Accordingly, the VA construction *sore=wa tanosi-i* in (54) is fundamentally different from the NA construction *sore=wa yukai-de aru* in (51), in that the former is schematically representable as the atomic proposition $\text{ENJOYABLE}(\text{sore})$ — whose default negation pattern is total, namely,

$\neg$ ENJOYABLE(**sore**) — whereas the latter can be represented as the conjunctive proposition **EXIST(sore) $\wedge$ PLEASANT(sore)** — whose default negation pattern is scope-limited, or partial, namely **EXIST(sore) $\wedge$ $\neg$ PLEASANT(sore)**.

It follows from the above discussion that the stem allomorph *at-* of the verb *ar-u* occurring in the past form *tanosi-k-at-ta*, for example, does not instantiate the one-argument existential verb *ar-* 'exist' used in nominal-adjectival constructions. Rather, it is an allomorph of the light verb stem *ar-* inserted as a matter of formal necessity in order to concatenate the verbal past-tense suffix *-ta* with the non-verbal VA stem *tanosi-ku*, as in *tanosi-ku + -ta* $\rightarrow$ *tanosi-ku + ar-* + *-ta* $\rightarrow$ *tanosi-k-at-ta*, as already discussed in Section 3. Thus, the syntactic derivation of the verbal-adjectival and nominal-adjectival constructions in (57)–(59) can be represented as follows:

(57) *Sore=wa tanosi-i*. 'It is enjoyable.'

Core construction

[*tanosi-ku*]

$\rightarrow$ [**sore** *tanosi-ku*]

$\rightarrow$ [[*sore tanosi-ku*] $T^0\emptyset$]

$\rightarrow$ [[[*sore tanosi-ku*] $T^0\emptyset$] **Top** $^0\emptyset$]

$\rightarrow$ [**sore_i=wa** [[[*t_i tanosi-ku*] $T^0\emptyset$] **Top** $^0\emptyset$]]

$\rightarrow$ *sore=wa tanosi-i* (allomorphy of *-ku* or amalgamation of *-ku* with $T^0\emptyset$ [–PAST])

$T^0\emptyset$ is an alternative notation for the phonologically null inflexional suffix $-\emptyset$ specified as [–PAST], already used in this paper. On the other hand, the category Topic, abbreviated as **Top**, is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme **Top** $^0\emptyset$, represented as $-\emptyset$. The element that moves to the specifier of **TopP** is assigned the particle *=wa* and functions to indicate the topic of the sentence.

In the non-past form in (56), when the underlying predication structure based on the VA stem *tanosi-ku* is combined with the non-past $T^0\emptyset$, the sequence [*-ku + T* $^0\emptyset$] is morphophonologically realised as *-i*, yielding the SF *tanosi-i*. The exponent *-i* may therefore be analysed either as a contextual allomorph *-i* of the VA thematic suffix *-ku* in the non-past environment or, alternatively, as an amalgamated realisation of the sequence [*-ku + T* $^0\emptyset$], as stated in Section 5.

(58) *Sore=wa tanosi-k-at-ta*. 'It was enjoyable.'

Core construction

[*tanosi-ku*]

→ [*sore tanosi-ku*]

→ [[*sore tanosi-ku*] **-ta**]

→ [[*sore tanosi-ku ar-*] **-ta**] (light-verb-stem insertion)

→ [[[*sore tanosi-ku ar-*] **-ta**] **Top⁰∅**]

→ [*sore_i=wa* [[[*t_i tanosi-ku ar-*] **-ta**] **Top⁰∅**]]]

→ *sore=wa tanosi-k+at-ta* (vowel deletion and consonant assimilation)

In the analysis of the VA past form in (58), the choice between the non-past **T⁰∅**, realised as **-∅**, and the past suffix **-ta** is treated as a paradigmatic selection made at a single derivational stage, since the **[-PAST]** **-∅** and the **[+PAST]** **-ta** constitute a paradigm whose terms acquire linguistic value solely through their opposition at the same point of speech. When the past suffix **-ta** is selected, it proves incompatible with non-verbal predicates such as *tanosi-ku*, thereby triggering the insertion of the light verb stem *ar-*.

In contrast to the verbal-adjectival constructions in (57) and (58), the nominal-adjectival construction in (59) consists of a core construction based on the VP headed by the one-argument existential verb stem *ar-* with an adjunctive ConjP based on the NAP headed by an NA stem:

(59) *Sore=wa yukai-de aru*. '(lit.) It exists, being pleasant.'

Adjunct

Core construction

[*yukai-ni*]

[*ar-*]

→ [*PRO yukai-ni*]

→ [*sore ar-*]

→ [[*PRO yukai-ni*] **-te**]

→ [[[*PRO_i yukai-ni*] **-te**] [*sore_i ar-*]] (adjunct embedding)

→ [[[[*PRO_i yukai-ni*] **-te**] [*sore_i ar-*]] **-ru**]

→ [[[[[[*PRO_i yukai-ni*] **-te**] [*sore_i ar-*]] **-ru**] **Top⁰∅**]

→ [*sore_i=wa* [[[[[[*PRO_i yukai-ni*] **-te**] [*t_i ar-*]] **-ru**] **Top⁰∅**]]]

→ *sore=wa yukai-de ar-u*

9. Analysis summary — Conclusion

(i) The morphosyntactic structure of the inflected forms of verbal adjectives is as shown in (60), where 'lit.' and 'col.' abbreviate 'literary form' and 'colloquial form', respectively:

(60)

Underlying form				Surface form
Compound stem: SENTENTIAL			Ending	
VA stem: LEXICAL		Light verb stem	Inflex. suff.	
VA rad.	VA them. suff.			
<i>taka-</i>	<i>-ku</i>	—	/ [0 PAST]	<i>taka-ku</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-ru</i> [–PAST]	<i>taka-i-Ø</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-ru</i> [–PAST] + NP	<i>taka-ki-Ø#NP</i> (lit.) <i>taka-i-Ø#NP</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-reba</i> 'if'	<i>taka-ke-reba</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-te</i>	<i>taka-ku-te</i> <i>taka-ku-tte</i> (col.)
<i>taka-</i>	<i>-ku</i>	—	<i>-temo</i>	<i>taka-ku-temo</i> <i>taka-ku-ttemo</i> (col.)
<i>taka-</i>	<i>-ku</i>	—	<i>-tatte</i>	<i>taka-ku-tatte</i> <i>taka-ku-ttatte</i> (col.)
<i>taka-</i>	<i>-ku</i>	(#ar-Ø)	<i>-tar-Ø</i> [+PAST]	<i>taka-k+at-ta</i>
<i>taka-</i>	<i>-ku</i>	(#ar-Ø)	<i>-tar-Ø-yō</i>	<i>taka-k+at-tar-ō</i>
<i>taka-</i>	<i>-ku</i>	(#ar-Ø)	<i>-tar-Ø-ba</i>	<i>taka-k+at-tar-a(-ba)</i>
<i>taka-</i>	<i>-ku</i>	(#ar-Ø)	<i>-tar-Ø-i</i>	<i>taka-k+at-tar-i</i>
<i>taka-</i>	<i>-ku</i>	(#ar-Ø)	<i>-yō</i>	<i>taka-k+ar-ō</i>
<i>taka-</i>	<i>-ku</i>	(#ar-Ø)	<i>-re</i>	<i>taka-k+ar-e</i>

Following the schematic representations in (38), the left-hand column lists the UFs and the right-hand column lists the corresponding SFs, so as to make explicit the allomorphy involved in the derivation. For uniform treatment with verbs, and as in the EMJ analysis in Section 6, the UF corresponding to the verbal-adjectival non-past suffix $-\emptyset$ is initially set to *-ru*, as in *taka-ku-ru* 'be high/tall', which is realised as *taka-i-Ø*. The underlying representations of the inflexional suffixes *-ta* and *-ō* are analysed as *-tar* and *-yō*, respectively. The suffixes *-te*, *-temo* and *-tatte* may be realised as *-tte*, *-ttemo* and *-ttatte* in colloquial registers, with emphatic consonantal gemination.

(ii) The VA radical is the element that encodes the lexical meaning specific to each verbal adjective. It can occur independently as a constituent of derived or compound words — for example, *taka-* in *taka-sa* 'height', *taka-mi* 'loftiness', *taka+nozomi* 'high hope' and *taka+dai* 'high platform' — but it cannot, on its own, be used as a syntactic constituent belonging to the category VA at the sentential level — for instance, *kanozyo=wa se=ga *taka* 'she is tall'; *asoko=ni se=no *taka otoko=ga iru* 'there is a tall man over there'; *koko=kara *taka kyodaina tō=ga mieru* 'from

here, a tower that, being tall, is colossal can be seen'; *watasi=wa nedan=ga *taka-reba kawanai* 'I do not buy it if the price is high'.

(iii) The function of the VA thematic suffix *-ku* $\approx$ *-i* $\approx$ *-(k)i* $\approx$ *-ke* is to licence the VA radical as a syntactic constituent of the category VA within sentential phrase structure. As noted in (ii), *taka-* on its own cannot be used as a VA in a sentence. Only when it combines with *-i*, *-(k)i*, *-ku* or *-ke* — forming *taka-i*, *taka-(k)i*, *taka-ku* or *taka-ke*, for example — does it become syntactically eligible to function as a constituent of the category VA — e.g. *kanozyo=wa se=ga takai*; *asoko=ni se=no takai otoko=ga iru*; *koko=kara takaku kyodaina tō=ga mieru*; *watasi=wa nedan=ga takake-reba kawanai*.

(iv) The form in which the VA thematic suffix occurs — *-i*, *-(k)i*, *-ku* or *-ke* — is determined by the syntactic environment in which the VA stem occurs, as shown in (61)–(64):

(61) When a VAP headed by a VA stem serves as the complement of a TP headed by the inflexional suffix *-reba*: ***-ku* $\rightarrow$ *-ke***

(62) When a VAP headed by a VA stem serves as the complement of a TP headed by $\emptyset$ [–PAST], and that TP is embedded within an NP: ***-ku* $\rightarrow$ *-(k)i***

(63) When a VAP headed by a VA stem serves as the complement of a TP headed by $\emptyset$ [–PAST], and that TP is not embedded within an NP: ***-ku* $\rightarrow$ *-i***

(64) In all other cases: ***-ku***

(v) At the surface level, *-ku* exhibits the phonological alternation $[-ku] \approx [-k]$, with vowel deletion occurring only when the VA stem combines with the stem *ar-* of the light verb *ar-u* or one of its allomorphs, resulting in the form $[-k]$. This vowel deletion can be taken to indicate that the VA stem in *-ku* and the light verb stem *ar-* have come to form a single compound stem at the sentential level.

(vi) It is also possible to interpret *-i* and *-(k)i* as synchronic amalgams of the VA thematic suffix and the tense suffix [–PAST]. Under this analysis, *-i* and *-(k)i* are not allomorphs exclusively of the VA thematic suffix but simultaneously encode [–PAST], thereby partially overlapping with the traditional analysis.

(vii) As shown in Section 6, there are significant structural parallels between the inflected forms of verbal adjectives — e.g. *takai* and its EMJ form *takasi* — and those of verbs, especially vowel-

stem verbs — e.g. *nobiru* and its EMJ form *nobu*. Here, the vowel-stem *k*-irregular verb *Kuru* 'come' is further examined, and the morphosyntactic structure of its inflected forms, semantically corresponding to those of the verbal adjective *takai* exemplified in (60), is shown in (65). Again, a strong parallelism in morphosyntactic structure can be observed:

(65)

Underlying form			Surface form
V stem		Ending	
V radical	V thematic suffix	Inflectional suffix(es)	
<i>K-</i>	<i>-i</i>	<i>-ru</i> [-PAST]	<i>K-u-ru</i>
<i>K-</i>	<i>-i</i>	<i>-reba</i>	<i>K-u-reba</i>
<i>K-</i>	<i>-i</i>	/ [0 PAST]	<i>K-i</i>
<i>K-</i>	<i>-i</i>	<i>-te</i>	<i>K-i-te</i>
<i>K-</i>	<i>-i</i>	<i>-temo</i>	<i>K-i-temo</i>
<i>K-</i>	<i>-i</i>	<i>-tatte</i>	<i>K-i-tatte</i>
<i>K-</i>	<i>-i</i>	<i>-tar-Ø</i> [+PAST]	<i>K-i-ta</i>
<i>K-</i>	<i>-i</i>	<i>-tar-Ø-yō</i>	<i>K-i-tar-ō</i>
<i>K-</i>	<i>-i</i>	<i>-tar-Ø-ba</i>	<i>K-i-tar-a(-ba)</i>
<i>K-</i>	<i>-i</i>	<i>-tar-Ø-i</i>	<i>K-i-tar-i</i>
<i>K-</i>	<i>-i</i>	<i>-yō</i>	<i>K-o-yō</i>
<i>K-</i>	<i>-i</i>	<i>-yo</i>	<i>K-o-i</i>

In vowel-stem verbs, the stem-final vowels — *-u* $\approx$ *-i* $\approx$ *-o* in the inflected forms of the verb [[[*K*]*u*]*ru*] as well as *-i* in those of the verb [[[*nob*]*i*]*ru*] in (36) and *-u* $\approx$ *-i* in those of the corresponding EMJ verb [[[*nob*]*u*] $\emptyset$] in (37) — can be understood as functioning as allomorphs of the V thematic suffix. For further details on the morphosyntactic structure of verbs, see the Appendix.

(viii) As discussed in Section 7, in the surface negative forms of verbal adjectives such as *takaku#nai-Ø*, *takaku#nak+at-ta*, *takaku#nak+ar-ō* and *takaku#nake-reba*, the constituents that are morphologically autonomous words, such as *nai-Ø*, *nak+at-ta*, *nak+ar-ō* and *nake-reba*, are assumed to be derived from intermediate strings such as **ara-nai-Ø*, **ara-nak+at-ta*, **ara-nak+ar-ō* and **ara-nake-reba*, due to the obligatory suppletion of the negative forms of the light verb *ar-u* by the inflected forms of the VA *nai*. Thus, the hypothetical intermediate forms corresponding to the above surface forms are **[[[takaku#ara]nai]Ø]*, **[[[takaku#ara]nak+at]ta]*, **[[[takaku#ara]nak+ar]ō]* and **[[[takaku#ara]nake]reba]*, which exhibit the same structure — that is, **[[[verbal stem]suffixal negator]ending]** — as verbal negative forms such as *[[[tabe]nai]Ø]*, *[[[tabe]nak+at]ta]*, *[[[tabe]nak+ar]ō]* and *[[[tabe]nake]reba]*, apart from the difference in structural complexity between the composite stem **[takaku#ara-]* and the simplex stem *[tabe-]*.

(ix) NA sentences consist syntactically of an existence-introducing, one-argument obligatory construction [$NP_i = wa \ t_i \ ar\text{-}T^0\text{-}Top^0\emptyset$] combined with a state-specifying adjunctive ConjP [$PRO_i \ NA_radical\text{-}ni\text{-}te$] and can be schematically represented as the conjunctive proposition $EXIST(NP) \wedge NA\text{-}STATE(NP)$, whose default negation pattern is scope-limited, namely $EXIST(NP) \wedge \neg NA\text{-}STATE(NP)$, whereas VA sentences consist only of a state-specifying obligatory construction [$NP_i = wa \ t_i \ VA_radical\text{-}ku\text{-}T^0\text{-}Top^0\emptyset$] without any adjunct and can be represented as the atomic proposition $VA\text{-}STATE(NP)$, whose default negation pattern is total, namely $\neg VA\text{-}STATE(NP)$. Thus, *at-* in the past NA construction *sore = wa sizuka-d-at-ta* '(lit.) it, being quiet/silent, existed', derived from [$sore_i = wa \ [PRO_i \ sizuka\text{-}ni\text{-}te] \ t_i \ ar\text{-}tar\text{-}Top^0\emptyset$], is an allomorph of the one-argument existential verb stem *ar-*, selected underlyingly. By contrast, *at-* in the past VA construction *sore = wa takak+at-ta* 'it was high/tall', derived from [$sore_i = wa \ t_i \ takaku\text{-}ar\text{-}ta\text{-}Top^0\emptyset$] is an allomorph of the light verb stem *ar-*, inserted due to the incompatibility between the non-verbal stem *takaku-* and the purely verbal suffix *-ta*. This syntactic difference is reflected in their negative forms, as in *sore = wa sizuka-de = wa/sizuka-d-ya nak+at-ta* '(lit.) it, not being quiet/silent, existed', with the particle *=wa* or its contextually restricted colloquial allomorph *-ya* marking the scope of negation, schematically representable as $EXIST(it) \wedge \neg QUIET(it)$, and *sore = wa takaku nak+at-ta* 'it was not high/tall', without this scope-limiting particle, schematically representable as $\neg HIGH(it)$.

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Appendix — Computation of Japanese inflected verbal-adjectival, true-verbal, nominal-adjectival and pseudo-copular forms from underlying representations

This appendix presents, with a view to possible applications, a fairly classical computation of both inflected verbal-adjectival forms and inflected true-verbal forms from their underlying representations. Its purpose is not to provide an exhaustive account of all verbal-adjectival and true-verbal morphology in Japanese, but to make it explicit that a substantial set of surface forms (SFs) of both categories can be derived by a single set of partially ordered operations, without positing separate derivations for verbal adjectives and true verbs. In this respect, the appendix is intended as a supplementary demonstration of the underlying structural parallelism between verbal adjectives and true verbs argued for in the main text — that is, between "**VA radical + VA thematic suffix *-ku* + inflexional suffix(es)**" and "**V radical + V thematic suffix $-\emptyset$ /*-i*/*-e* + inflexional suffix(es)**". The SFs can be generated from the underlying representations by the ordered application of the following operations. The appendix also presents a computation of inflected nominal-adjectival forms and pseudo-copular forms, most of which are to be analysed as "**NA radical + NA thematic suffix *-ni* + adjunctive suffix *-te***" followed by "**V radical *ar-* + V thematic suffix $-\emptyset$ + inflexional suffix(es)**" and "**NP + postposition *=ni* + adjunctive suffix *-te***" followed by "**V radical *ar-* + V thematic suffix $-\emptyset$ + inflexional suffix(es)**", respectively. Note crucially that, in the operations below, * represents a wildcard that also covers #NP, rather than ungrammaticality.

Partially ordered operations

1. Change ***-ru**** to **$-\emptyset$ *** after ***-ku***.

Optionally, change ***-ru**** to **$-\emptyset$ *** after ***-ni-te#ar- $\emptyset$*** .

Optionally, change ***-ru*** to **$-\emptyset$** after ***=ni-te#ar- $\emptyset$*** except before **#NP**.

Note that ***-ni*** and ***=ni*** are distinct morphemes.

2. Change ***-ku*** to ***-i*** before **$-\emptyset$ ***.

Optionally, change ***-ku*** to ***-ki*** before **$-\emptyset$ #NP**, except after ***ōki***, in a more literary register.

Change ***-ku*** to ***-ke*** before ***-reba***.

3. Optionally, change ***yo*** to ***i*** before ***-i- $\emptyset$ **** in a more colloquial register.

4. Insert ***#ar- $\emptyset$*** after ***-ku*** before ***-tar****, ***-yō***, ***-re***, ***-na****, ***-zu***, ***-mai***, ***/-ta****, ***/-tutu***, ***/-nagara***, ***/-gatera***.

5. Insert ***-ru*** between **$-\emptyset$** and ***-mai***.

Optionally, insert ***-ru*** between ***-i*** or ***-e*** and ***-mai***.

6. Before ***-na****, ***-zu*** and ***-ba***, change **$\emptyset$** to ***a***.

- Before **-ru** and **-reba**, change *i* to *u* after **S-** and **K-** in the **s-irregular** and **k-irregular** verbs.
- Before **-yo** and **-zu**, change *i* to *e* after **S-** in the **s-irregular** verb.
- Before **-sas*** and **-rar***, change *i* to $\emptyset$ or to *e* after **S-** in the **s-irregular** verb.
- Before **-yō**, **-yo**, **-na***, **-zu**, **-sas***, **-rar*** and **-mai**, change *i* to *o* after **K-** in the **k-irregular** verb.
- Optionally, before **-mai**, change *i* to *u* after **S-** in the **s-irregular** verb.
- Optionally, change *e* to *u* in the verbs **Ø-e-ru** 'gain' and **p-e-ru** — phonetically [heru] — 'pass (time)' before **-ru** and **-reba** except before **-ru-mai**.
7. Delete **-te** following **-ni** before **#ar-Ø-Ø#NP**, **#ar-Ø-Ø=no**, **#ar-a-ba** and **#ar-a-zu**.
Delete **-te** following **=ni** before **#ar-Ø-Ø=no**, **#ar-a-ba** and **#ar-a-zu**.
Optionally, delete **-te** following **-ni** before **#ar-Ø-ru#NP** and **#ar-Ø-re** in a more literary register.
 8. Change **-ni-te**, **=ni-te**, **-ni-temo**, **=ni-temo**, **-ni-tatte** and **=ni-tatte** to **-de**, **=de**, **-demo**, **=demo**, **-datte** and **=datte** respectively.
 9. Change **-ni#ar-** to **-n-ar-**.
Change **=ni#ar-** to **=n-ar-** except before **a-zu**.
 10. Optionally, delete **-ba** after **-n-ar-a**, **=n-ar-a** and **-tar-a** in a more contemporary register.
 11. Delete all remaining instances of **-Ø** and **Ø-**.
 12. Insert *i* immediately after a consonant before */*.
 13. Delete */*.
 14. Delete suffix-initial *s*, *r* and *y* after any consonant.
 15. Change **-re** to **-ro** in all verbs and **-yo** to **-i** in the **k-irregular** verb.
 16. Before **-te**, **-temo**, **-tatte**, **-tar***, change *ow* to *ō*, obligatorily in *to-u* 'ask (a question)', underlyingly *tow-Ø-ru*, and *ko-u* 'beg', underlyingly *kow-Ø-ru*, and optionally in *ito-u* 'dislike', underlyingly *itow-Ø-ru*, and *iko-u* 'rest, relax', underlyingly *ikow-Ø-ru*.
 17. Before **-te**, **-temo**, **-tatte**, **-tar***, change *r* and *w* to *t*.
Before **-te**, **-temo**, **-tatte**, **-tar****, change *s* to *si*.
Before **-te**, **-temo**, **-tatte**, **-tar***, change *k* to *i*, except in the verb *ik-u* 'go', underlyingly *ik-Ø-ru*, where *k* changes to *t*.
 18. In **-te**, **-temo**, **-tatte**, **-tar***, change suffix-initial *t* to *d* after ***b**, ***g**, ***m**, ***n**.
 19. Before **-de**, **-demo**, **-datte**, **-dar***, change *b* and *m* to *n*.

Before *-de*, *-demo*, *-datte*, *-dar**, change *g* to *i*.

20. Delete all *w* except before *a**.

21. Change *-ku#at* to *-k+at* before *-tar**.

Change *-ku#ar* to *-k+ar* before *-ō*, *-e* and *-a-zu*.

Optionally, change *-ku#ar* to *-k+ar* before *-u-mai* in literary register.

22. Change *-de#* and *=de#* to *-d-* and *=d-*, respectively, before *ar*, when *ar* is word-final.

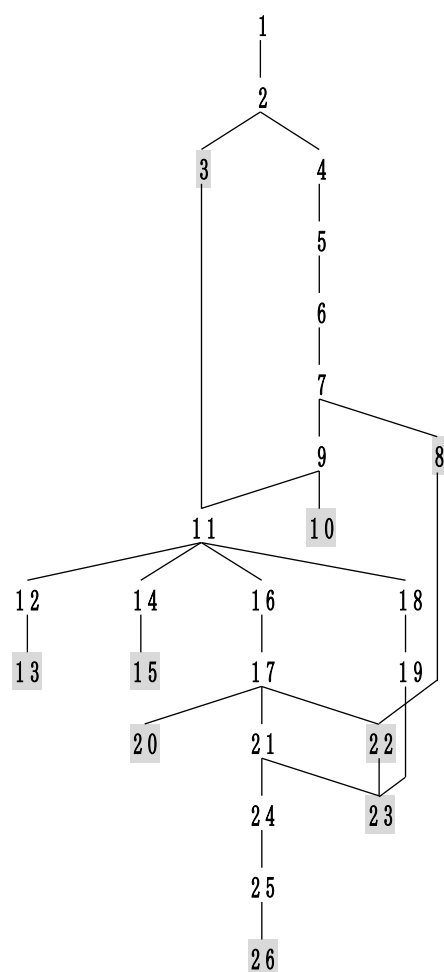
Optionally, change *-de#* and *=de#* to *-d-* and *=d-*, respectively, before *at-tar** and *ar-ō* in a slightly less formal register.

23. Delete word-final *r*. (Note that # and = both indicate word boundaries.)

24. Delete *ar-a-* before *-na**.

25. Insert *=wa* between *-de* and *=de*, on the one hand, and *#na** and *#ar-u-mai*, on the other.

26. Optionally, change *-de=wa* and *=de=wa* to *-dy-a* and *=dy-a* in a more colloquial register.



Ordering

1 → 2 → {3, (4 → 5 → 6 → 7)} in either order → {8, 9} in either order → {10, 11} in either order {(12 → 13), (14 → 15), (16 → 17), (18 → 19)} in any order → {20, 21, 22} in any order → {23, (24 → 25 → 26)} in either order. More precisely, 3 may apply at any point between 2 and 11; 8 at any point between 7 and 22; 10 at any point after 9; 13 at any point after 12; 15 at any point after 14; 20 at any point after 17; 22 at any point after both 8 and 17 and before 23; 23 at any point after 19, 21 and 22; and 26 at any point after 25. Note that, **in practice, the SFs are generated by applying operations 1 through 26 sequentially.**

In the schematic representation below, consonant-stem verbs have a verbal stem whose underlying V thematic suffix is the phonologically null *-Ø*, as in *kak-Ø-ru* 'write', realised as *kak-u*, whereas vowel-stem verbs have a verbal stem whose underlying V

thematic suffix is *-i* or *-e*, as in *k-i-ru* 'wear' and *tab-e-ru* 'eat'.

Underlying form			Surface form
Stem		Ending	
Radical	Thematic suffix	Inflexional or derivational ⁵ suffix(es)	
<i>taka</i>	<i>-ku</i>	<i>-ru</i> [–PAST]	<i>taka-i</i> 'be high/tall'
<i>taka</i>	<i>-ku</i>	<i>-ru#NP</i> [–PAST]	<i>taka-i#NP</i> <i>taka-ki#NP</i> (lit.)
<i>taka</i>	<i>-ku</i>	<i>-reba</i>	<i>taka-ke-reba</i>
<i>taka</i>	<i>-ku</i>	/ [0 PAST]	<i>taka-ku</i>
<i>taka</i>	<i>-ku</i>	/-ta-ku	<i>taka-ku#ari-ta-ku</i> ⁶
<i>taka</i>	<i>-ku</i>	/-tutu	<i>taka-ku#ari-tutu</i>
<i>taka</i>	<i>-ku</i>	/-nagara	<i>taka-ku#ari-nagara</i>
<i>taka</i>	<i>-ku</i>	<i>-te</i>	<i>taka-ku-te</i>
<i>taka</i>	<i>-ku</i>	<i>-temo</i>	<i>taka-ku-temo</i>
<i>taka</i>	<i>-ku</i>	<i>-tatte</i>	<i>taka-ku-tatte</i>
<i>taka</i>	<i>-ku</i>	<i>-tar-Ø</i> [+PAST]	<i>taka-k+at-ta</i>
<i>taka</i>	<i>-ku</i>	<i>-tar-Ø-yō</i>	<i>taka-k+at-tar-ō</i>
<i>taka</i>	<i>-ku</i>	<i>-tar-Ø-ba</i>	<i>taka-k+at-tar-a(-ba)</i>
<i>taka</i>	<i>-ku</i>	<i>-tar-Ø-i</i>	<i>taka-k+at-tar-i</i>
<i>taka</i>	<i>-ku</i>	<i>-yō</i>	<i>taka-k+ar-ō</i> (lit.)
<i>taka</i>	<i>-ku</i>	<i>-re</i>	<i>taka-k+ar-e</i> (lit.)
<i>taka</i>	<i>-ku</i>	<i>-na-ku</i>	<i>taka-ku#na-ku</i>
<i>taka</i>	<i>-ku</i>	<i>-zu</i>	<i>taka-k+ar-a-zu</i> (lit.)
<i>taka</i>	<i>-ku</i>	<i>-mai</i>	<i>taka-ku#ar-u-mai</i> <i>taka-k+ar-u-mai</i> (lit.)

<i>yo</i>	<i>-ku</i>	<i>-ru</i> [–PAST]	<i>yo-i</i> 'be good'
<i>yo</i>	<i>-ku</i>	<i>-ru#NP</i> [–PAST]	<i>i-i</i> <i>yo-i#NP</i> <i>i-i#NP</i> <i>yo-ki#NP</i> (lit.)
<i>yo</i>	<i>-ku</i>	<i>-reba</i>	<i>yo-ke-reba</i>
<i>yo</i>	<i>-ku</i>	/ [0 PAST]	<i>yo-ku</i>
<i>yo</i>	<i>-ku</i>	/-ta-ku	<i>yo-ku#ari-ta-ku</i>
<i>yo</i>	<i>-ku</i>	/-tutu	<i>yo-ku#ari-tutu</i>
<i>yo</i>	<i>-ku</i>	/-nagara	<i>yo-ku#ari-nagara</i>
<i>yo</i>	<i>-ku</i>	<i>-te</i>	<i>yo-ku-te</i>
<i>yo</i>	<i>-ku</i>	<i>-temo</i>	<i>yo-ku-temo</i>
<i>yo</i>	<i>-ku</i>	<i>-tatte</i>	<i>yo-ku-tatte</i>
<i>yo</i>	<i>-ku</i>	<i>-tar-Ø</i> [+PAST]	<i>yo-k+at-ta</i>
<i>yo</i>	<i>-ku</i>	<i>-tar-Ø-yō</i>	<i>yo-k+at-tar-ō</i>
<i>yo</i>	<i>-ku</i>	<i>-tar-Ø-ba</i>	<i>yo-k+at-tar-a(-ba)</i>
<i>yo</i>	<i>-ku</i>	<i>-tar-Ø-i</i>	<i>yo-k+at-tar-i</i>

⁵ More precisely, derivational suffixes are not part of verbal endings.

⁶ Although some doubt remains as to whether *taka-ku#ari-ta-ku*, *taka-ku#ari-tutu* and *taka-ku#ari-nagara* should be treated as inflected forms of the VA *takai* — since *ari* in these forms may be the stem not of an inserted light verb but of a one-argument existential verb — they are included here in order to show that these forms, too, can be generated. The same holds of *yo-ku#ari-ta-ku*, *yo-ku#ari-tutu* and *yo-ku#ari-nagara* below.

<i>yo</i>	<i>-ku</i>	<i>-yō</i>
<i>yo</i>	<i>-ku</i>	<i>-re</i>
<i>yo</i>	<i>-ku</i>	<i>-na-ku</i>
<i>yo</i>	<i>-ku</i>	<i>-zu</i>
<i>yo</i>	<i>-ku</i>	<i>-mai</i>

<i>yo-k+ar-ō</i> (lit.)
<i>yo-k+ar-e</i> (lit.)
<i>yo-ku#na-ku</i>
<i>yo-k+ar-a-zu</i> (lit.)
<i>yo-ku#ar-u-mai</i>
<i>yo-k+ar-u-mai</i> (lit.)

<i>k</i>	<i>-i</i>	<i>-ru</i> [–PAST]
<i>k</i>	<i>-i</i>	<i>-reba</i>
<i>k</i>	<i>-i</i>	/ [0 PAST]
<i>k</i>	<i>-i</i>	/-tutu
<i>k</i>	<i>-i</i>	/-nagara
<i>k</i>	<i>-i</i>	/-gatera
<i>k</i>	<i>-i</i>	/-ta-ku
<i>k</i>	<i>-i</i>	<i>-te</i>
<i>k</i>	<i>-i</i>	<i>-temo</i>
<i>k</i>	<i>-i</i>	<i>-tatte</i>
<i>k</i>	<i>-i</i>	<i>-tar-Ø</i> [+PAST]
<i>k</i>	<i>-i</i>	<i>-tar-Ø-yō</i>
<i>k</i>	<i>-i</i>	<i>-tar-Ø-ba</i>
<i>k</i>	<i>-i</i>	<i>-tar-Ø-i</i>
<i>k</i>	<i>-i</i>	<i>-yō</i>
<i>k</i>	<i>-i</i>	<i>-re</i>
<i>k</i>	<i>-i</i>	<i>-yo</i>
<i>k</i>	<i>-i</i>	<i>-na-ku</i>
<i>k</i>	<i>-i</i>	<i>-zu</i>
<i>k</i>	<i>-i</i>	<i>-sas-e</i>
<i>k</i>	<i>-i</i>	<i>-rar-e</i>
<i>k</i>	<i>-i</i>	<i>-mai</i>

<i>k-i-ru</i> 'wear'
<i>k-i-reba</i>
<i>k-i</i>
<i>k-i-tutu</i>
<i>k-i-nagara</i>
<i>k-i-gatera</i>
<i>k-i-ta-ku</i>
<i>k-i-te</i>
<i>k-i-temo</i>
<i>k-i-tatte</i>
<i>k-i-ta</i>
<i>k-i-tar-ō</i>
<i>k-i-tar-a(-ba)</i>
<i>k-i-tar-i</i>
<i>k-i-yō</i>
<i>k-i-ro</i>
<i>k-i-yo</i> (lit.)
<i>k-i-na-ku</i>
<i>k-i-zu</i> (lit.)
<i>k-i-sas-e</i>
<i>k-i-rar-e</i>
<i>k-i-mai</i>
<i>k-i-ru-mai</i>

<i>tab</i>	<i>-e</i>	<i>-ru</i> [–PAST]
<i>tab</i>	<i>-e</i>	<i>-reba</i>
<i>tab</i>	<i>-e</i>	/ [0 PAST]
<i>tab</i>	<i>-e</i>	/-tutu
<i>tab</i>	<i>-e</i>	/-nagara
<i>tab</i>	<i>-e</i>	/-gatera
<i>tab</i>	<i>-e</i>	/-ta-ku
<i>tab</i>	<i>-e</i>	<i>-te</i>
<i>tab</i>	<i>-e</i>	<i>-temo</i>
<i>tab</i>	<i>-e</i>	<i>-tatte</i>
<i>tab</i>	<i>-e</i>	<i>-tar-Ø</i> [+PAST]
<i>tab</i>	<i>-e</i>	<i>-tar-Ø-yō</i>
<i>tab</i>	<i>-e</i>	<i>-tar-Ø-ba</i>
<i>tab</i>	<i>-e</i>	<i>-tar-Ø-i</i>
<i>tab</i>	<i>-e</i>	<i>-yō</i>
<i>tab</i>	<i>-e</i>	<i>-re</i>
<i>tab</i>	<i>-e</i>	<i>-yo</i>
<i>tab</i>	<i>-e</i>	<i>-na-ku</i>
<i>tab</i>	<i>-e</i>	<i>-zu</i>

<i>tab-e-ru</i> 'eat'
<i>tab-e-reba</i>
<i>tab-e</i>
<i>tab-e-tutu</i>
<i>tab-e-nagara</i>
<i>tab-e-gatera</i>
<i>tab-e-ta-ku</i>
<i>tab-e-te</i>
<i>tab-e-temo</i>
<i>tab-e-tatte</i>
<i>tab-e-ta</i>
<i>tab-e-tar-ō</i>
<i>tab-e-tar-a(-ba)</i>
<i>tab-e-tar-i</i>
<i>tab-e-yō</i>
<i>tab-e-ro</i>
<i>tab-e-yo</i> (lit.)
<i>tab-e-na-ku</i>
<i>tab-e-zu</i> (lit.)

<i>tab</i>	<i>-e</i>	<i>-sas-e</i>
<i>tab</i>	<i>-e</i>	<i>-rar-e</i>
<i>tab</i>	<i>-e</i>	<i>-mai</i>

<i>tab-e-sas-e</i>
<i>tab-e-rar-e</i>
<i>tab-e-mai</i>
<i>tab-e-ru-mai</i>

∅	<i>-e</i>	<i>-ru</i> [–PAST]
∅	<i>-e</i>	<i>-reba</i>
∅	<i>-e</i>	/ [0 PAST]
∅	<i>-e</i>	<i>/-tutu</i>
∅	<i>-e</i>	<i>/-nagara</i>
∅	<i>-e</i>	<i>/-gatera</i>
∅	<i>-e</i>	<i>/-ta-ku</i>
∅	<i>-e</i>	<i>-te</i>
∅	<i>-e</i>	<i>-temo</i>
∅	<i>-e</i>	<i>-tatte</i>
∅	<i>-e</i>	<i>-tar-∅</i> [+PAST]
∅	<i>-e</i>	<i>-tar-∅-yō</i>
∅	<i>-e</i>	<i>-tar-∅-ba</i>
∅	<i>-e</i>	<i>-tar-∅-i</i>
∅	<i>-e</i>	<i>-yō</i>
∅	<i>-e</i>	<i>-re</i>
∅	<i>-e</i>	<i>-yo</i>
∅	<i>-e</i>	<i>-na-ku</i>
∅	<i>-e</i>	<i>-zu</i>
∅	<i>-e</i>	<i>-sas-e</i>
∅	<i>-e</i>	<i>-rar-e</i>
∅	<i>-e</i>	<i>-mai</i>

<i>e-ru</i> 'gain'
<i>u-ru</i> (lit.)
<i>e-reba</i>
<i>u-reba</i> (lit.)
<i>e</i>
<i>e-tutu</i>
<i>e-nagara</i>
<i>e-gatera</i>
<i>e-ta-ku</i>
<i>e-te</i>
<i>e-temo</i>
<i>e-tatte</i>
<i>e-ta</i>
<i>e-tar-ō</i>
<i>e-tar-a(-ba)</i>
<i>e-tar-i</i>
<i>e-yō</i>
<i>e-ro</i>
<i>e-yo</i> (lit.)
<i>e-na-ku</i>
<i>e-zu</i> (lit.)
<i>e-sas-e</i>
<i>e-rar-e</i>
<i>e-mai</i>
<i>e-ru-mai</i>

<i>K</i> ⁷	<i>-i</i>	<i>-ru</i> [–PAST]
<i>K</i>	<i>-i</i>	<i>-reba</i>
<i>K</i>	<i>-i</i>	/ [0 PAST]
<i>K</i>	<i>-i</i>	<i>/-tutu</i>
<i>K</i>	<i>-i</i>	<i>/-nagara</i>
<i>K</i>	<i>-i</i>	<i>/-gatera</i>
<i>K</i>	<i>-i</i>	<i>/-ta-ku</i>
<i>K</i>	<i>-i</i>	<i>-te</i>
<i>K</i>	<i>-i</i>	<i>-temo</i>
<i>K</i>	<i>-i</i>	<i>-tatte</i>
<i>K</i>	<i>-i</i>	<i>-tar-∅</i> [+PAST]
<i>K</i>	<i>-i</i>	<i>-tar-∅-yō</i>
<i>K</i>	<i>-i</i>	<i>-tar-∅-ba</i>
<i>K</i>	<i>-i</i>	<i>-tar-∅-i</i>

<i>K-u-ru</i> 'come' (<i>k</i> -irregular)
<i>K-u-reba</i>
<i>K-i</i>
<i>K-i-tutu</i>
<i>K-i-nagara</i>
<i>K-i-gatera</i>
<i>K-i-ta-ku</i>
<i>K-i-te</i>
<i>K-i-temo</i>
<i>K-i-tatte</i>
<i>K-i-ta</i>
<i>K-i-tar-ō</i>
<i>K-i-tar-a(-ba)</i>
<i>K-i-tar-i</i>

⁷ In order to distinguish the *k*-irregular vowel-stem verb *k-u-ru* 'come' from the regular vowel-stem verb *k-i-ru* 'wear', whose radical is also /k-/, the former radical /k-/ will be notated as capital *K*- in this appendix, as in *K-u-ru*. Similarly, the radical of the *s*-irregular vowel-stem verb *s-u-ru* 'do' will be designated as capital *S*-, as in *S-u-ru*.

<i>K</i>	<i>-i</i>	<i>-yō</i>
<i>K</i>	<i>-i</i>	<i>-yo</i>
<i>K</i>	<i>-i</i>	<i>-na-ku</i>
<i>K</i>	<i>-i</i>	<i>-zu</i>
<i>K</i>	<i>-i</i>	<i>-sas-e</i>
<i>K</i>	<i>-i</i>	<i>-rar-e</i>
<i>K</i>	<i>-i</i>	<i>-mai</i>

<i>K-o-yō</i>
<i>K-o-i</i>
<i>K-o-na-ku</i>
<i>K-o-zu</i>
<i>K-o-sas-e</i>
<i>K-o-rar-e</i>
<i>K-o-mai</i> (lit.)
<i>K-u-ru-mai</i>

<i>S</i>	<i>-i</i>	<i>-ru</i> [–PAST]
<i>S</i>	<i>-i</i>	<i>-reba</i>
<i>S</i>	<i>-i</i>	/ [0 PAST]
<i>S</i>	<i>-i</i>	/-tutu
<i>S</i>	<i>-i</i>	/-nagara
<i>S</i>	<i>-i</i>	/-gatera
<i>S</i>	<i>-i</i>	/-ta-ku
<i>S</i>	<i>-i</i>	<i>-te</i>
<i>S</i>	<i>-i</i>	<i>-temo</i>
<i>S</i>	<i>-i</i>	<i>-tatte</i>
<i>S</i>	<i>-i</i>	<i>-tar-Ø</i> [+PAST]
<i>S</i>	<i>-i</i>	<i>-tar-Ø-yō</i>
<i>S</i>	<i>-i</i>	<i>-tar-Ø-ba</i>
<i>S</i>	<i>-i</i>	<i>-tar-Ø-i</i>
<i>S</i>	<i>-i</i>	<i>-yō</i>
<i>S</i>	<i>-i</i>	<i>-re</i>
<i>S</i>	<i>-i</i>	<i>-yo</i>
<i>S</i>	<i>-i</i>	<i>-na-ku</i>
<i>S</i>	<i>-i</i>	<i>-zu</i>
<i>S</i>	<i>-i</i>	<i>-sas-e</i>
<i>S</i>	<i>-i</i>	<i>-rar-e</i>
<i>S</i>	<i>-i</i>	<i>-mai</i>

<i>S-u-ru</i> 'do' (s-irregular)
<i>S-u-reba</i>
<i>S-i</i>
<i>S-i-tutu</i>
<i>S-i-nagara</i>
<i>S-i-gatera</i>
<i>S-i-ta-ku</i>
<i>S-i-te</i>
<i>S-i-temo</i>
<i>S-i-tatte</i>
<i>S-i-ta</i>
<i>S-i-tar-ō</i>
<i>S-i-tar-a(-ba)</i>
<i>S-i-tar-i</i>
<i>S-i-yō</i>
<i>S-i-ro</i>
<i>S-e-yo</i> (lit.)
<i>S-i-na-ku</i>
<i>S-e-zu</i> (lit.)
<i>S-as-e</i>
<i>S-e-sas-e</i> (lit.)
<i>S-ar-e</i>
<i>S-e-rar-e</i> (lit.)
<i>S-i-mai</i> (lit.)
<i>S-u-mai</i> (lit.)
<i>S-u-ru-mai</i>

<i>kes</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>kes</i>	<i>-Ø</i>	<i>-reba</i>
<i>kes</i>	<i>-Ø</i>	/ [0 PAST]
<i>kes</i>	<i>-Ø</i>	/-tutu
<i>kes</i>	<i>-Ø</i>	/-nagara
<i>kes</i>	<i>-Ø</i>	/-gatera
<i>kes</i>	<i>-Ø</i>	/-ta-ku
<i>kes</i>	<i>-Ø</i>	<i>-te</i>
<i>kes</i>	<i>-Ø</i>	<i>-temo</i>
<i>kes</i>	<i>-Ø</i>	<i>-tatte</i>
<i>kes</i>	<i>-Ø</i>	<i>-tar-Ø</i> [+PAST]
<i>kes</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>
<i>kes</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>
<i>kes</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>
<i>kes</i>	<i>-Ø</i>	<i>-yō</i>

<i>kes-u</i> 'erase, delete'
<i>kes-eba</i>
<i>kesi</i>
<i>kesi-tutu</i>
<i>kesi-nagara</i>
<i>kesi-gatera</i>
<i>kesi-ta-ku</i>
<i>kesi-te</i>
<i>kesi-temo</i>
<i>kesi-tatte</i>
<i>kesi-ta</i>
<i>kesi-tar-ō</i>
<i>kesi-tar-a(-ba)</i>
<i>kesi-tar-i</i>
<i>kes-ō</i>

<i>kes</i>	<i>-Ø</i>	<i>-re</i>
<i>kes</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>kes</i>	<i>-Ø</i>	<i>-zu</i>
<i>kes</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>kes</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>kes</i>	<i>-Ø</i>	<i>-mai</i>

<i>kes-e</i>
<i>kes-a-na-ku</i>
<i>kes-a-zu</i> (lit.)
<i>kes-as-e</i>
<i>kes-ar-e</i>
<i>kes-u-mai</i>

<i>kak</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>kak</i>	<i>-Ø</i>	<i>-reba</i>
<i>kak</i>	<i>-Ø</i>	/ [0 PAST]
<i>kak</i>	<i>-Ø</i>	/-tutu
<i>kak</i>	<i>-Ø</i>	/-nagara
<i>kak</i>	<i>-Ø</i>	/-gatera
<i>kak</i>	<i>-Ø</i>	/-ta-ku
<i>kak</i>	<i>-Ø</i>	<i>-te</i>
<i>kak</i>	<i>-Ø</i>	<i>-temo</i>
<i>kak</i>	<i>-Ø</i>	<i>-tatte</i>
<i>kak</i>	<i>-Ø</i>	<i>-tar-Ø</i> [+PAST]
<i>kak</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>
<i>kak</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>
<i>kak</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>
<i>kak</i>	<i>-Ø</i>	<i>-yō</i>
<i>kak</i>	<i>-Ø</i>	<i>-re</i>
<i>kak</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>kak</i>	<i>-Ø</i>	<i>-zu</i>
<i>kak</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>kak</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>kak</i>	<i>-Ø</i>	<i>-mai</i>

<i>kak-u</i> 'write'
<i>kak-eba</i>
<i>kaki</i>
<i>kaki-tutu</i>
<i>kaki-nagara</i>
<i>kaki-gatera</i>
<i>kaki-ta-ku</i>
<i>kai-te</i>
<i>kai-temo</i>
<i>kai-tatte</i>
<i>kai-ta</i>
<i>kai-tar-ō</i>
<i>kai-tar-a(-ba)</i>
<i>kai-tar-i</i>
<i>kak-ō</i>
<i>kak-e</i>
<i>kak-a-na-ku</i>
<i>kak-a-zu</i> (lit.)
<i>kak-as-e</i>
<i>kak-ar-e</i>
<i>kak-u-mai</i>

<i>ik</i>	<i>-Ø</i>	<i>-ru</i> [–PAST]
<i>ik</i>	<i>-Ø</i>	<i>-reba</i>
<i>ik</i>	<i>-Ø</i>	/ [0 PAST]
<i>ik</i>	<i>-Ø</i>	/-tutu
<i>ik</i>	<i>-Ø</i>	/-nagara
<i>ik</i>	<i>-Ø</i>	/-gatera
<i>ik</i>	<i>-Ø</i>	/-ta-ku
<i>ik</i>	<i>-Ø</i>	<i>-te</i>
<i>ik</i>	<i>-Ø</i>	<i>-temo</i>
<i>ik</i>	<i>-Ø</i>	<i>-tatte</i>
<i>ik</i>	<i>-Ø</i>	<i>-tar-Ø</i> [+PAST]
<i>ik</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>
<i>ik</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>
<i>ik</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>
<i>ik</i>	<i>-Ø</i>	<i>-yō</i>
<i>ik</i>	<i>-Ø</i>	<i>-re</i>
<i>ik</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>ik</i>	<i>-Ø</i>	<i>-zu</i>
<i>ik</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>ik</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>ik</i>	<i>-Ø</i>	<i>-mai</i>

<i>ik-u</i> 'go'
<i>ik-eba</i>
<i>iki</i>
<i>iki-tutu</i>
<i>iki-nagara</i>
<i>iki-gatera</i>
<i>iki-ta-ku</i>
<i>it-te</i>
<i>it-temo</i>
<i>it-tatte</i>
<i>it-ta</i>
<i>it-tar-ō</i>
<i>it-tar-a(-ba)</i>
<i>it-tar-i</i>
<i>ik-ō</i>
<i>ik-e</i>
<i>ik-a-na-ku</i>
<i>ik-a-zu</i> (lit.)
<i>ik-as-e</i>
<i>ik-ar-e</i>
<i>ik-u-mai</i>

<i>nug</i>	-Ø	-ru [-PAST]
<i>nug</i>	-Ø	-reba
<i>nug</i>	-Ø	/ [0 PAST]
<i>nug</i>	-Ø	/-tutu
<i>nug</i>	-Ø	/-nagara
<i>nug</i>	-Ø	/-gatera
<i>nug</i>	-Ø	/-ta-ku
<i>nug</i>	-Ø	-te
<i>nug</i>	-Ø	-temo
<i>nug</i>	-Ø	-tatte
<i>nug</i>	-Ø	-tar-Ø [+PAST]
<i>nug</i>	-Ø	-tar-Ø-yō
<i>nug</i>	-Ø	-tar-Ø-ba
<i>nug</i>	-Ø	-tar-Ø-i
<i>nug</i>	-Ø	-yō
<i>nug</i>	-Ø	-re
<i>nug</i>	-Ø	-na-ku
<i>nug</i>	-Ø	-zu
<i>nug</i>	-Ø	-sas-e
<i>nug</i>	-Ø	-rar-e
<i>nug</i>	-Ø	-mai

<i>nug-u</i> 'take off (clothes)'
<i>nug-eba</i>
<i>nugi</i>
<i>nugi-tutu</i>
<i>nugi-nagara</i>
<i>nugi-gatera</i>
<i>nugi-ta-ku</i>
<i>nui-de</i>
<i>nui-demo</i>
<i>nui-datte</i>
<i>nui-da</i>
<i>nui-dar-ō</i>
<i>nui-dar-a(-ba)</i>
<i>nui-dar-i</i>
<i>nug-ō</i>
<i>nug-e</i>
<i>nug-a-na-ku</i>
<i>nug-a-zu</i> (lit.)
<i>nug-as-e</i>
<i>nug-ar-e</i>
<i>nug-u-mai</i>

<i>ut</i>	-Ø	-ru [-PAST]
<i>ut</i>	-Ø	-reba
<i>ut</i>	-Ø	/ [0 PAST]
<i>ut</i>	-Ø	/-tutu
<i>ut</i>	-Ø	/-nagara
<i>ut</i>	-Ø	/-gatera
<i>ut</i>	-Ø	/-ta-ku
<i>ut</i>	-Ø	-te
<i>ut</i>	-Ø	-temo
<i>ut</i>	-Ø	-tatte
<i>ut</i>	-Ø	-tar-Ø [+PAST]
<i>ut</i>	-Ø	-tar-Ø-yō
<i>ut</i>	-Ø	-tar-Ø-ba
<i>ut</i>	-Ø	-tar-Ø-i
<i>ut</i>	-Ø	-yō
<i>ut</i>	-Ø	-re
<i>ut</i>	-Ø	-na-ku
<i>ut</i>	-Ø	-zu
<i>ut</i>	-Ø	-sas-e
<i>ut</i>	-Ø	-rar-e
<i>ut</i>	-Ø	-mai

<i>ut-u</i> 'hit, beat'
<i>ut-eba</i>
<i>uti</i>
<i>uti-tutu</i>
<i>uti-nagara</i>
<i>uti-gatera</i>
<i>uti-ta-ku</i>
<i>ut-te</i>
<i>ut-temo</i>
<i>ut-tatte</i>
<i>ut-ta</i>
<i>ut-tar-ō</i>
<i>ut-tar-a(-ba)</i>
<i>ut-tar-i</i>
<i>ut-ō</i>
<i>ut-e</i>
<i>ut-a-na-ku</i>
<i>ut-a-zu</i> (lit.)
<i>ut-as-e</i>
<i>ut-ar-e</i>
<i>ut-u-mai</i>

<i>sin</i>	-Ø	-ru [-PAST]
<i>sin</i>	-Ø	-reba
<i>sin</i>	-Ø	/ [0 PAST]
<i>sin</i>	-Ø	/-tutu
<i>sin</i>	-Ø	/-nagara
<i>sin</i>	-Ø	/-gatera

<i>sin-u</i> 'die'
<i>sin-eba</i>
<i>sini</i>
<i>sini-tutu</i>
<i>sini-nagara</i>
<i>sini-gatera</i>

<i>sin</i>	<i>-Ø</i>	<i>/-ta-ku</i>
<i>sin</i>	<i>-Ø</i>	<i>-te</i>
<i>sin</i>	<i>-Ø</i>	<i>-temo</i>
<i>sin</i>	<i>-Ø</i>	<i>-tatte</i>
<i>sin</i>	<i>-Ø</i>	<i>-tar-Ø [+PAST]</i>
<i>sin</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>
<i>sin</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>
<i>sin</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>
<i>sin</i>	<i>-Ø</i>	<i>-yō</i>
<i>sin</i>	<i>-Ø</i>	<i>-re</i>
<i>sin</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>sin</i>	<i>-Ø</i>	<i>-zu</i>
<i>sin</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>sin</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>sin</i>	<i>-Ø</i>	<i>-mai</i>

<i>sini-ta-ku</i>
<i>sin-de</i>
<i>sin-demo</i>
<i>sin-datte</i>
<i>sin-da</i>
<i>sin-dar-ō</i>
<i>sin-dar-a(-ba)</i>
<i>sin-dar-i</i>
<i>sin-ō</i>
<i>sin-e</i>
<i>sin-a-na-ku</i>
<i>sin-a-zu</i>
<i>sin-as-e</i>
<i>sin-ar-e</i>
<i>sin-u-mai</i>

<i>kir</i>	<i>-Ø</i>	<i>-ru [-PAST]</i>
<i>kir</i>	<i>-Ø</i>	<i>-reba</i>
<i>kir</i>	<i>-Ø</i>	<i>/ [0 PAST]</i>
<i>kir</i>	<i>-Ø</i>	<i>/-tutu</i>
<i>kir</i>	<i>-Ø</i>	<i>/-nagara</i>
<i>kir</i>	<i>-Ø</i>	<i>/-gatera</i>
<i>kir</i>	<i>-Ø</i>	<i>/-ta-ku</i>
<i>kir</i>	<i>-Ø</i>	<i>-te</i>
<i>kir</i>	<i>-Ø</i>	<i>-temo</i>
<i>kir</i>	<i>-Ø</i>	<i>-tatte</i>
<i>kir</i>	<i>-Ø</i>	<i>-tar-Ø [+PAST]</i>
<i>kir</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>
<i>kir</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>
<i>kir</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>
<i>kir</i>	<i>-Ø</i>	<i>-yō</i>
<i>kir</i>	<i>-Ø</i>	<i>-re</i>
<i>kir</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>kir</i>	<i>-Ø</i>	<i>-zu</i>
<i>kir</i>	<i>-Ø</i>	<i>-sas-e</i>
<i>kir</i>	<i>-Ø</i>	<i>-rar-e</i>
<i>kir</i>	<i>-Ø</i>	<i>-mai</i>

<i>kir-u 'cut'</i>
<i>kir-eba</i>
<i>kiri</i>
<i>kiri-tutu</i>
<i>kiri-nagara</i>
<i>kiri-gatera</i>
<i>kiri-ta-ku</i>
<i>kit-te</i>
<i>kit-temo</i>
<i>kit-tatte</i>
<i>kit-ta</i>
<i>kit-tar-ō</i>
<i>kit-tar-a(-ba)</i>
<i>kit-tar-i</i>
<i>kir-ō</i>
<i>kir-e</i>
<i>kir-a-na-ku</i>
<i>kir-a-zu (lit.)</i>
<i>kir-as-e</i>
<i>kir-ar-e</i>
<i>kir-u-mai</i>

<i>ar</i>	<i>-Ø</i>	<i>-ru [-PAST]</i>
<i>ar</i>	<i>-Ø</i>	<i>-reba</i>
<i>ar</i>	<i>-Ø</i>	<i>/ [0 PAST]</i>
<i>ar</i>	<i>-Ø</i>	<i>/-tutu</i>
<i>ar</i>	<i>-Ø</i>	<i>/-nagara</i>
<i>ar</i>	<i>-Ø</i>	<i>/-gatera</i>
<i>ar</i>	<i>-Ø</i>	<i>/-ta-ku</i>
<i>ar</i>	<i>-Ø</i>	<i>-te</i>
<i>ar</i>	<i>-Ø</i>	<i>-temo</i>
<i>ar</i>	<i>-Ø</i>	<i>-tatte</i>
<i>ar</i>	<i>-Ø</i>	<i>-tar-Ø [+PAST]</i>
<i>ar</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>

<i>ar-u 'exist'</i>
<i>ar-eba</i>
<i>ari</i>
<i>ari-tutu</i>
<i>ari-nagara</i>
<i>ari-gatera</i>
<i>ari-ta-ku</i>
<i>at-te</i>
<i>at-temo</i>
<i>at-tatte</i>
<i>at-ta</i>
<i>at-tar-ō</i>

<i>ar</i>	-Ø	<i>-tar-Ø-ba</i>
<i>ar</i>	-Ø	<i>-tar-Ø-i</i>
<i>ar</i>	-Ø	<i>-yō</i>
<i>ar</i>	-Ø	<i>-re</i>
<i>ar</i>	-Ø	<i>-na-ku</i>
<i>ar</i>	-Ø	<i>-zu</i>
<i>ar</i>	-Ø	<i>-sas-e</i>
<i>ar</i>	-Ø	<i>-rar-e</i>
<i>ar</i>	-Ø	<i>-mai</i>

<i>at-tar-a(-ba)</i>
<i>at-tar-i</i>
<i>ar-ō</i>
<i>ar-e</i>
<i>na-ku</i> (suppletion)
<i>ar-a-zu</i> (lit.)
<i>ar-as-e</i>
<i>ar-ar-e</i>
<i>ar-u-mai</i>

<i>tob</i>	-Ø	<i>-ru</i> [–PAST]
<i>tob</i>	-Ø	<i>-reba</i>
<i>tob</i>	-Ø	/ [0 PAST]
<i>tob</i>	-Ø	<i>/-tutu</i>
<i>tob</i>	-Ø	<i>/-nagara</i>
<i>tob</i>	-Ø	<i>/-gatera</i>
<i>tob</i>	-Ø	<i>/-ta-ku</i>
<i>tob</i>	-Ø	<i>-te</i>
<i>tob</i>	-Ø	<i>-temo</i>
<i>tob</i>	-Ø	<i>-tatte</i>
<i>tob</i>	-Ø	<i>-tar-Ø</i> [+PAST]
<i>tob</i>	-Ø	<i>-tar-Ø-yō</i>
<i>tob</i>	-Ø	<i>-tar-Ø-ba</i>
<i>tob</i>	-Ø	<i>-tar-Ø-i</i>
<i>tob</i>	-Ø	<i>-yō</i>
<i>tob</i>	-Ø	<i>-re</i>
<i>tob</i>	-Ø	<i>-na-ku</i>
<i>tob</i>	-Ø	<i>-zu</i>
<i>tob</i>	-Ø	<i>-sas-e</i>
<i>tob</i>	-Ø	<i>-rar-e</i>
<i>tob</i>	-Ø	<i>-mai</i>

<i>tob-u</i> 'fly'
<i>tob-eba</i>
<i>tobi</i>
<i>tobi-tutu</i>
<i>tobi-nagara</i>
<i>tobi-gatera</i>
<i>tobi-ta-ku</i>
<i>ton-de</i>
<i>ton-demo</i>
<i>ton-datte</i>
<i>ton-da</i>
<i>ton-dar-ō</i>
<i>ton-dar-a(-ba)</i>
<i>ton-dar-i</i>
<i>tob-ō</i>
<i>tob-e</i>
<i>tob-a-na-ku</i>
<i>tob-a-zu</i> (lit.)
<i>tob-as-e</i>
<i>tob-ar-e</i>
<i>tob-u-mai</i>

<i>yom</i>	-Ø	<i>-ru</i> [–PAST]
<i>yom</i>	-Ø	<i>-reba</i>
<i>yom</i>	-Ø	/ [0 PAST]
<i>yom</i>	-Ø	<i>/-tutu</i>
<i>yom</i>	-Ø	<i>/-nagara</i>
<i>yom</i>	-Ø	<i>/-gatera</i>
<i>yom</i>	-Ø	<i>/-ta-ku</i>
<i>yom</i>	-Ø	<i>-te</i>
<i>yom</i>	-Ø	<i>-temo</i>
<i>yom</i>	-Ø	<i>-tatte</i>
<i>yom</i>	-Ø	<i>-tar-Ø</i> [+PAST]
<i>yom</i>	-Ø	<i>-tar-Ø-yō</i>
<i>yom</i>	-Ø	<i>-tar-Ø-ba</i>
<i>yom</i>	-Ø	<i>-tar-Ø-i</i>
<i>yom</i>	-Ø	<i>-yō</i>
<i>yom</i>	-Ø	<i>-re</i>
<i>yom</i>	-Ø	<i>-na-ku</i>
<i>yom</i>	-Ø	<i>-zu</i>

<i>yom-u</i> 'read'
<i>yom-eba</i>
<i>yomi</i>
<i>yomi-tutu</i>
<i>yomi-nagara</i>
<i>yomi-gatera</i>
<i>yomi-ta-ku</i>
<i>yon-de</i>
<i>yon-demo</i>
<i>yon-datte</i>
<i>yon-da</i>
<i>yon-dar-ō</i>
<i>yon-dar-a(-ba)</i>
<i>yon-dar-i</i>
<i>yom-ō</i>
<i>yom-e</i>
<i>yom-a-na-ku</i>
<i>yom-a-zu</i> (lit.)

<i>yom</i>	-Ø	-sas-e
<i>yom</i>	-Ø	-rar-e
<i>yom</i>	-Ø	-mai

<i>yom-as-e</i>
<i>yom-ar-e</i>
<i>yom-u-mai</i>

<i>kaw</i>	-Ø	-ru [-PAST]
<i>kaw</i>	-Ø	-reba
<i>kaw</i>	-Ø	/ [0 PAST]
<i>kaw</i>	-Ø	/-tutu
<i>kaw</i>	-Ø	/-nagara
<i>kaw</i>	-Ø	/-gatera
<i>kaw</i>	-Ø	/-ta-ku
<i>kaw</i>	-Ø	-te
<i>kaw</i>	-Ø	-temo
<i>kaw</i>	-Ø	-tatte
<i>kaw</i>	-Ø	-tar-Ø [+PAST]
<i>kaw</i>	-Ø	-tar-Ø-yō
<i>kaw</i>	-Ø	-tar-Ø-ba
<i>kaw</i>	-Ø	-tar-Ø-i
<i>kaw</i>	-Ø	-yō
<i>kaw</i>	-Ø	-re
<i>kaw</i>	-Ø	-na-ku
<i>kaw</i>	-Ø	-zu
<i>kaw</i>	-Ø	-sas-e
<i>kaw</i>	-Ø	-rar-e
<i>kaw</i>	-Ø	-mai

<i>ka-u</i> 'buy'
<i>ka-eba</i>
<i>kai</i>
<i>kai-tutu</i>
<i>kai-nagara</i>
<i>kai-gatera</i>
<i>kai-ta-ku</i>
<i>kat-te</i>
<i>kat-temo</i>
<i>kat-tatte</i>
<i>kat-ta</i>
<i>kat-tar-ō</i>
<i>kat-tar-a(-ba)</i>
<i>kat-tar-i</i>
<i>ka-ō</i>
<i>ka-e</i>
<i>kaw-a-na-ku</i>
<i>kaw-a-zu</i>
<i>kaw-as-e</i>
<i>kaw-ar-e</i>
<i>ka-u-mai</i>

<i>tow</i>	-Ø	-ru [-PAST]
<i>tow</i>	-Ø	-reba
<i>tow</i>	-Ø	/ [0 PAST]
<i>tow</i>	-Ø	/-tutu
<i>tow</i>	-Ø	/-nagara
<i>tow</i>	-Ø	/-gatera
<i>tow</i>	-Ø	/-ta-ku
<i>tow</i>	-Ø	-te
<i>tow</i>	-Ø	-temo
<i>tow</i>	-Ø	-tatte
<i>tow</i>	-Ø	-tar-Ø [+PAST]
<i>tow</i>	-Ø	-tar-Ø-yō
<i>tow</i>	-Ø	-tar-Ø-ba
<i>tow</i>	-Ø	-tar-Ø-i
<i>tow</i>	-Ø	-yō
<i>tow</i>	-Ø	-re
<i>tow</i>	-Ø	-na-ku
<i>tow</i>	-Ø	-zu
<i>tow</i>	-Ø	-sas-e
<i>tow</i>	-Ø	-rar-e
<i>tow</i>	-Ø	-mai

<i>to-u</i> 'ask (question)'
<i>to-eba</i>
<i>toi</i>
<i>toi-tutu</i>
<i>toi-nagara</i>
<i>toi-gatera</i>
<i>toi-ta-ku</i>
<i>tō-te</i>
<i>tō-temo</i>
<i>tō-tatte</i>
<i>tō-ta</i>
<i>tō-tar-ō</i>
<i>tō-tar-a(-ba)</i>
<i>tō-tar-i</i>
<i>to-ō</i>
<i>to-e</i>
<i>tow-a-na-ku</i>
<i>tow-a-zu</i> (lit.)
<i>tow-as-e</i>
<i>tow-ar-e</i>
<i>to-u-mai</i>

<i>ow</i>	-Ø	-ru [-PAST]
<i>ow</i>	-Ø	-reba

<i>o-u</i> 'chase'
<i>o-eba</i>

<i>ow</i>	<i>-Ø</i>	<i>/ [0 PAST]</i>	<i>oi</i>
<i>ow</i>	<i>-Ø</i>	<i>/-tutu</i>	<i>oi-tutu</i>
<i>ow</i>	<i>-Ø</i>	<i>/-nagara</i>	<i>oi-nagara</i>
<i>ow</i>	<i>-Ø</i>	<i>/-gatera</i>	<i>oi-gatera</i>
<i>ow</i>	<i>-Ø</i>	<i>/-ta-ku</i>	<i>oi-ta-ku</i>
<i>ow</i>	<i>-Ø</i>	<i>-te</i>	<i>ot-te</i>
<i>ow</i>	<i>-Ø</i>	<i>-temo</i>	<i>ot-temo</i>
<i>ow</i>	<i>-Ø</i>	<i>-tatte</i>	<i>ot-tatte</i>
<i>ow</i>	<i>-Ø</i>	<i>-tar-Ø [+PAST]</i>	<i>ot-ta</i>
<i>ow</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>	<i>ot-tar-ō</i>
<i>ow</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>	<i>ot-tar-a(-ba)</i>
<i>ow</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>	<i>ot-tar-i</i>
<i>ow</i>	<i>-Ø</i>	<i>-yō</i>	<i>o-ō</i>
<i>ow</i>	<i>-Ø</i>	<i>-re</i>	<i>o-e</i>
<i>ow</i>	<i>-Ø</i>	<i>-na-ku</i>	<i>ow-a-na-ku</i>
<i>ow</i>	<i>-Ø</i>	<i>-zu</i>	<i>ow-a-zu</i> (lit.)
<i>ow</i>	<i>-Ø</i>	<i>-sas-e</i>	<i>ow-as-e</i>
<i>ow</i>	<i>-Ø</i>	<i>-rar-e</i>	<i>ow-ar-e</i>
<i>ow</i>	<i>-Ø</i>	<i>-mai</i>	<i>o-u-mai</i>

The following are causative, passive, causative-passive and potential verb forms:

Underlying form							Surface form
Stem: SENTENTIAL						Ending	
Stem: LEXICAL		Stem: LEXICAL		Stem: LEXICAL		Infl.suff.	
Rad.	Them.suff.	Rad.	Them.suff.	Rad.	Them.suff.		
<i>tab</i>	<i>-e</i>	<i>-sas</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>tab-e-sas-e-ru</i>
<i>tab</i>	<i>-e</i>	<i>-sas</i>	<i>-Ø</i>	—	—	<i>-ru</i>	<i>tab-e-sas-u</i>
<i>nom</i>	<i>-Ø</i>	<i>-sas</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>nom-as-e-ru</i>
<i>nom</i>	<i>-Ø</i>	<i>-sas</i>	<i>-Ø</i>	—	—	<i>-ru</i>	<i>nom-as-u</i>
<i>tab</i>	<i>-e</i>	<i>-rar</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>tab-e-rar-e-ru</i>
<i>nom</i>	<i>-Ø</i>	<i>-rar</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>nom-ar-e-ru</i>
<i>tab</i>	<i>-e</i>	<i>-sas</i>	<i>-e</i>	<i>-rar</i>	<i>-e</i>	<i>-ru</i>	<i>tab-e-sas-e-rar-e-ru</i>
<i>nom</i>	<i>-Ø</i>	<i>-sas</i>	<i>-Ø</i>	<i>-rar</i>	<i>-e</i>	<i>-ru</i>	<i>nom-as-ar-e-ru</i>
<i>tab</i>	<i>-e</i>	<i>-rar</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>tab-e-rar-e-ru</i>
<i>nom</i>	<i>-Ø</i>	<i>-r</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>tab-e-r-e-ru</i> (coll.)
<i>nom</i>	<i>-Ø</i>	<i>-r</i>	<i>-e</i>	—	—	<i>-ru</i>	<i>nom-e-ru</i>

The morphosyntactic structures of NA forms and pseudo-copular forms are more complex, as shown below, where 'Them.suf.', 'Conj.suf.', 'Inflex.suf.' and 'Postpos.' stand for 'thematic suffix', 'conjunctive suffix', 'inflexional suffix(es)' and 'postposition', respectively:

Underlying form						Surface form
Stem		Ending	Stem		Ending	
Radical	Them.suf.	Conj.suf.	Radical	Them.suf.	Inflex.suf.	
<i>sizuka</i>	<i>-ni</i>	—				<i>sizuka-ni</i> 'quiet; silent'
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>				<i>sizuka-de</i>
<i>sizuka</i>	<i>-ni</i>	<i>-temo</i>				<i>sizuka-demo</i>
<i>sizuka</i>	<i>-ni</i>	<i>-tatte</i>				<i>sizuka-datte</i>

<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ru</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ru#NP</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ru=no</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-reba</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ba</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/-tutu</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/-nagara</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/-ta-ku</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-te</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-temo</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tatte</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-yō</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-re</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-na-ku</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-zu</i>
<i>sizuka</i>	<i>-ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-mai</i>

<i>sizuka-d-a</i>
<i>sizuka-de#ar-u</i>
<i>sizuka-n-a#NP</i>
<i>sizuka-n-ar-u#NP</i>
<i>sizuka-de#ar-u#NP</i>
<i>sizuka-n-a=no</i>
<i>sizuka-de#ar-u=no</i>
<i>sizuka-de#ar-eba</i>
<i>sizuka-n-ar-a(-ba)</i>
<i>sizuka-de#ari</i>
<i>sizuka-de#ari-tutu</i>
<i>sizuka-de#ari-nagara</i>
<i>sizuka-de#ari-ta-ku</i>
<i>sizuka-de#at-te</i>
<i>sizuka-de#at-temo</i>
<i>sizuka-de#at-tatte</i>
<i>sizuka-d-at-ta</i>
<i>sizuka-de#at-ta</i>
<i>sizuka-d-at-tar-ō</i>
<i>sizuka-de#at-tar-ō</i>
<i>sizuka-d-at-tar-a(-ba)</i>
<i>sizuka-de#at-tar-a(-ba)</i>
<i>sizuka-d-at-tar-i</i>
<i>sizuka-de#at-tar-i</i>
<i>sizuka-d-ar-ō</i>
<i>sizuka-de#ar-ō</i>
<i>sizuka-n-ar-e (lit.)</i>
<i>sizuka-de#ar-e</i>
<i>sizuka-de=wa#na-ku</i>
<i>sizuka-dy-a#na-ku</i>
<i>sizuka-n-ar-a-zu</i>
<i>sizuka-de=wa#ar-u-mai</i>
<i>sizuka-dy-a#ar-u-mai</i>

Underlying form					
Conjunction phrase			Stem		Ending
NP	Postpos.	Conj.suf.	Radical	Them.suf.	Inflex.suf.
<i>gakusei</i> 'student'	<i>=ni</i>	<i>-te</i>			
<i>gakusei</i>	<i>=ni</i>	<i>-temo</i>			
<i>gakusei</i>	<i>=ni</i>	<i>-tatte</i>			
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ru</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ru#NP</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ru=no</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-reba</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-ba</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/-tutu</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/-nagara</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>/-ta-ku</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-te</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-temo</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tatte</i>

Surface form
<i>gakusei=de</i>
<i>gakusei=demo</i>
<i>gakusei=datte</i>
<i>gakusei=d-a</i>
<i>gakusei=de#ar-u</i>
<i>gakusei=de#ar-u#NP</i>
<i>gakusei=n-a=no</i>
<i>gakusei=de#ar-u=no</i>
<i>gakusei=de#ar-eba</i>
<i>gakusei=n-ar-a(-ba)</i>
<i>gakusei=de#ari</i>
<i>gakusei=de#ari-tutu</i>
<i>gakusei=de#ari-nagara</i>
<i>gakusei=de#ari-ta-ku</i>
<i>gakusei=de#at-te</i>
<i>gakusei=de#at-temo</i>
<i>gakusei=de#at-tatte</i>

<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø</i>	<i>gakusei=d-at-ta</i> <i>gakusei=de#at-ta</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø-yō</i>	<i>gakusei=d-at-tar-ō</i> <i>gakusei=de#at-tar-ō</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø-ba</i>	<i>gakusei=d-at-tar-a(-ba)</i> <i>gakusei=de#at-tar-a(-ba)</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-tar-Ø-i</i>	<i>gakusei=d-at-tar-i</i> <i>gakusei=de#at-tar-i</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-yō</i>	<i>gakusei=d-ar-ō</i> <i>gakusei=de#ar-ō</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-re</i>	<i>gakusei=de#ar-e</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-na-ku</i>	<i>gakusei=de=wa#na-ku</i> <i>gakusei=dy-a#na-ku</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-zu</i>	<i>gakusei=ni#ar-a-zu</i>
<i>gakusei</i>	<i>=ni</i>	<i>-te</i>	<i>#ar</i>	<i>-Ø</i>	<i>-mai</i>	<i>gakusei=de=wa#ar-u-mai</i> <i>gakusei=dy-a#ar-u-mai</i>

The interpretation of the morphophonological operations described above is as follows:

1. Alternation of *-ru* with *-Ø* after *-ku*, *-ni-te#ar-Ø* and *=ni-te#ar-Ø*

After the VA thematic suffix *-ku*, the non-past suffix *-ru** alternates obligatorily with *-Ø**. Thus:

taka-ku-ru → *taka-ku-Ø*
taka-ku-ru#NP → *taka-ku-Ø#NP*
tab-e-na-ku-ru → *tab-e-na-ku-Ø*

This process shows that, after *-ku*, the tense suffix [–PAST] is realised as phonologically null *-Ø*, thereby creating the environment for the subsequent alternations of *-ku* with *-i* or *-ki* described in 2 below. Furthermore, in a more colloquial register, the tense suffix *-ru* is frequently realised as *-Ø* after *-ni-te#ar-Ø* in NA forms and, in pseudo-copular forms, after *=ni-te#ar-Ø*, except before an NP. Thus:

sizuka-ni-te#ar-Ø-ru → *sizuka-ni-te#ar-Ø-ru* / *sizuka-ni-te#ar-Ø-Ø*
sizuka-ni-te#ar-Ø-ru#NP → *sizuka-ni-te#ar-Ø-ru#NP* / *sizuka-ni-te#ar-Ø-Ø#NP*
gakusei=ni-te#ar-Ø-ru → *gakusei=ni-te#ar-Ø-ru* / *gakusei=ni-te#ar-Ø-Ø*
gakusei=ni-te#ar-Ø-ru#NP → *gakusei=ni-te#ar-Ø-ru#NP*⁸

⁸ Instead of the form *gakusei=ni-te#ar-Ø-ru#NP*, which surfaces as *gakusei=de#ar-u#NP*, as in *gakusei=de#ar-u wakamono* 'a young person who is a student', another structurally distinct form,

In particular, *-ni-te#ar-Ø-Ø#NP* with the covert *-Ø* in the second row above — surfacing as *-n-a#NP*, as in *sizuka-n-a#heya* 'a room that is quiet' — is, even in written language, the modern standard form, rather than *-ni-te#ar-Ø-ru#NP* with the overt *-ru*, which surfaces as *-n-ar-u#NP* in a more literary register, as in *sizuka-n-ar-u#heya*, or as *-de#ar-u#NP*, as in *sizuka-de#ar-u#heya*, the latter of which appears not to be very common⁹.

2. Alternations of *-ku* with *-i* $\approx$ *-(k)i* $\approx$ *-ke* before *-Ø**, *-Ø#NP* and *-reba*

Before the non-past suffix *-Ø**, the VA thematic suffix *-ku* alternates with the allomorph *-i*. Optionally, before *-Ø#NP*, *-ku* alternates with *-ki* in a more literary register. Before the conditional suffix *-reba*, *-ku* alternates with *-ke*. Thus:

taka-ku-Ø → *taka-i-Ø*¹⁰

taka-ku-Ø#NP → *taka-i-Ø#NP* / *taka-ki-Ø#NP*

taka-ku-reba → *taka-ke-reba*

tab-e-na-ku-Ø → *tab-e-na-i-Ø*

tab-e-na-ku-reba → *tab-e-na-ke-reba*

These alternations indicate that *-ku* has distinct allomorphs depending on the following environment: before *-Ø**, the regular outcome is *-i*; before *-Ø#NP*, *-ki* is also possible in a more literary register, with the exception of the non-existent form **ōki-ki-Ø#NP* from the VA *ōkii* 'be big'; and before *-reba*, the allomorph *-ke* occurs.

gakusei=no#NP with the genitive particle *=no*, may be used, as in *gakusei=no wakamono* 'a young person of student status'.

⁹ There are a few NA radicals, such as *ōki-* and *ōi-*, that are lexically specified as having idiosyncratic SFs, as in *ōki-n-a#NP* (but not **ōki-n-ar-u#NP* or **ōki-de#ar-u#NP*), derived from *ōki-ni-te#ar-Ø-Ø#NP*, and *ōi-n-ar-u#NP* (but not **ōi-n-a#NP* or **ōi-de#ar-u#NP*), derived from *ōi-ni-te#ar-Ø-ru#NP*. Moreover, *ōki-ni-te#ar-Ø-Ø* and *ōi-ni-te#ar-Ø-ru* can be used only within NP-modifying relative clauses. The former type of NAs, traditionally called *rentaisi*, includes *ōki-n-a* 'big', *tīsa-n-a* 'small', *okasi-n-a* 'funny; strange', *iron-n-a* 'various' and *roku-n-a* 'decent; proper', whereas the latter type, which belongs to a more literary register, includes *ōi-n-ar-u* 'great', *tan-n-ar-u* 'mere; simple', *ika-n-a-ru* 'what kind of; any', *sara-n-a-ru* 'further; additional', *sei-n-ar-u* 'holy; sacred' and *haha-n-ar-u* 'motherly; maternal'.

¹⁰ In a highly literary register, in genres such as *waka*, *haiku* and *senryū*, the form *taka-si-Ø* remains possible today, although the corresponding form **ōki-si-Ø* 'be big' is non-existent for historical reasons. This classical allomorph *-si* of the VA thematic suffix *-ku* is realised as *-Ø* after a VA radical ending in *-si*, as in *utukusi-ku-Ø* 'be beautiful' → *utukusi-i-Ø* (standard) / *utukusi-Ø-Ø* (highly literary), rather than **utukusi-si-Ø*.

3. Allomorphy of *yo-* as *i-* before *-i-Ø**

The radical *yo-* of the VA *yo-i* 'be good', which is underlyingly set as *yo-ku-ru* in this appendix, has a more colloquial allomorph *i-* only before the VA thematic-suffix allomorph *-i*, and not before the other allomorphs *-ku*, *-ki* or *-ke*. Thus:

<i>yo-i-Ø*</i>	→ <i>i-i-Ø*</i> / <i>yo-i-Ø*</i>
<i>yo-ki-Ø#NP</i>	→ <i>yo-ki-Ø#NP</i>
<i>yo-ku#</i>	→ <i>yo-ku#</i>
<i>yo-ke-reba</i>	→ <i>yo-ke-reba</i>

4. Insertion of *#ar-Ø* after *-ku*

Before the suffixes *-tar**, *-yō*, *-re*, *-na**, *-zu*, *-mai*, */-ta**, */-tutu* and */-nagara*, *#ar-Ø* is inserted after the VA thematic suffix *-ku*. Thus:

<i>taka-ku-tar-Ø</i>	→ <i>taka-ku#ar-Ø-tar-Ø</i>
<i>taka-ku-yō</i>	→ <i>taka-ku#ar-Ø-yō</i>
<i>taka-ku-re</i>	→ <i>taka-ku#ar-Ø-re</i>
<i>taka-ku-na-ku</i>	→ <i>taka-ku#ar-Ø-na-ku</i>
<i>taka-ku-zu</i>	→ <i>taka-ku#ar-Ø-zu</i>
<i>taka-ku-mai</i>	→ <i>taka-ku#ar-Ø-mai</i>
<i>taka-ku/-ta-ku</i>	→ <i>taka-ku#ar-Ø/-ta-ku</i>
<i>taka-ku/-tutu</i>	→ <i>taka-ku#ar-Ø/-tutu</i>
<i>taka-ku/-nagara</i>	→ <i>taka-ku#ar-Ø/-nagara</i>

This process indicates that these forms are not built directly on bare *-ku*. Rather, they are formed on a structure in which *#ar-Ø* is introduced after *-ku*. The verbal-adjectival system therefore distinguishes two types of configuration after *-ku*. In one type, represented by underlying forms (Ufs) such as *taka-ku-te*, *taka-ku-temo*, *taka-ku-tatte*, *taka-ku-Ø*, *taka-ku-Ø#NP* and *taka-ku-reba*, the element following *-ku* directly determines the surface allomorph of *-ku*, namely *-ku*, *-i*, *-ki* or *-ke*. In the other type, represented by UFs such as *taka-ku-tar**, *taka-ku-yō*, *taka-ku-re*, *taka-ku-na**, *taka-ku-zu*, *taka-ku-mai*, *taka-ku/-ta**, *taka-ku/-tutu* and *taka-ku/-nagara*, the relevant form

is not built directly on bare *-ku*, but on a structure containing the verbal stem *ar-Ø* after *-ku*. Accordingly, the opposition is not simply between different suffixes following *-ku*, but rather between forms built directly on *-ku* and forms built on *-ku* followed by *#ar-Ø*.

5. Insertion of *-ru* before *-mai*

Before the negative-volitional/assumptive suffix *-mai*, consonant-stem verbs show insertion of the non-past tense suffix *-ru* following the V thematic suffix *-Ø*. Thus:

kes-Ø-mai → *kes-Ø-**ru**-mai* (→ *kes-ru-mai* → *kes-u-mai*)

taka-ku#ar-Ø-mai → *taka-ku#ar-Ø-**ru**-mai* (→ *taka-ku#ar-ru-mai* → *taka-ku#ar-u-mai*)

This process indicates that *-mai* does not attach directly to the bare consonant-final V stem. Rather, the non-past suffix *-ru* is first introduced, after which the regular morphophonological operations then apply. In vowel-stem verbs, including the s-irregular verb *S-u-ru* 'do' and the k-irregular verb *K-u-ru* 'come' — here underlyingly set as *S-i-ru* and *K-i-ru*, respectively — the insertion of *-ru* following the V thematic suffix *-i* is also optionally possible. Thus:

tab-e-mai → *tab-e-**ru**-mai*

S-i-mai → *S-i-**ru**-mai* (→ *S-u-ru-mai*)

K-i-mai → *K-i-**ru**-mai* (→ *K-u-ru-mai*)

6. Verbal-stem alternations

Before the negative suffixes *-na** and *-zu* and the conditional suffix *-ba*, the V thematic suffix *-Ø* alternates with *-a*, yielding a special stem used only before these negative suffixes and this conditional suffix. Thus:

kes-Ø-na-ku → *kes-**a**-na-ku*

kes-Ø-zu → *kes-**a**-zu*

tab-e-tar-Ø-ba → *tab-e-tar-**a**-ba*

sizuka-ni#ar-Ø-zu → *sizuka-ni#ar-**a**-zu*

sizuka-ni#ar-Ø-ba → *sizuka-ni#ar-**a**-ba*

gakusei=ni#ar-Ø-zu → *gakusei=ni#ar-a-zu*

gakusei=ni#ar-Ø-ba → *gakusei=ni#ar-a-ba*

In the *s*-irregular verb *S-u-ru* and the *k*-irregular verb *K-u-ru*, the following alternations of the V thematic suffix are attested:

S-i → *S-u* / ___ -ru, -reba

S-i → *S-e* / ___ -zu, -yo

S-i → *S-Ø* / ___ -sas*, -rar*

S-i → *S-e* / ___ -sas*, -rar* (only in literary register)

K-i → *K-u* / ___ -ru, -reba

K-i → *K-o* / ___ -yō, -yo, -na*, -zu, -sas*, -rar*, -mai

In the *s*-irregular verb, moreover, the following alternation of the V thematic suffix *-i* is possible:

S-i → *S-u* / ___ -mai

Lastly, in the vowel-stem verbs *Ø-e-ru* 'gain' and *p-e-ru* [heru] 'pass (time)', the following alternations of the V thematic suffix are possible in a more literary register:

Ø-e → *Ø-u* / ___ -ru, -reba

p-e → *p-u* / ___ -ru, -reba

7. Deletion of *-te* after *-ni* and *=ni*

In a small number of NA forms such as *sizuka-n-a#NP*, *sizuka-n-ar-u#NP* (lit.), *sizuka-n-a=no*, *sizuka-n-ar-e* (lit.), *sizuka-n-ar-a(-ba)* and *sizuka-n-ar-a-zu* (lit.), the adjunctive suffix *-te* does not surface before the one-argument existential verb stem *ar-*. Thus, the underlying *-te* is assumed to be deleted obligatorily before *#ar-Ø-Ø#NP*, *#ar-Ø-Ø=no*¹¹, *#ar-a-ba* and *#ar-a-zu*, and optionally before *#ar-Ø-ru#NP* and *#ar-Ø-re*, as follows:

¹¹ As with the enclitic formal noun *=no*, the suffix *-te* is also deleted in *-ni-te#ar-Ø-Ø* before the nouns *#dake*, *#wake*, *#bacari* and *#mon*, as in *sizuka-n-a#dake#at-te*, *sizuka-n-a#wake=ga*, *sizuka-n-a#bacari=ni* and *sizuka-n-a#mon(o)=de*, derived from *sizuka-ni-te#ar-Ø-Ø#dake#ar-Ø-te*, *sizuka-ni-*

<i>sizuka-ni-te#ar-Ø-Ø#NP</i>	→ <i>sizuka-ni#ar-Ø-Ø#NP</i>
<i>sizuka-ni-te#ar-Ø-Ø=no</i>	→ <i>sizuka-ni#ar-Ø-Ø=no</i>
<i>sizuka-ni-te#ar-a-ba</i>	→ <i>sizuka-ni#ar-a-ba</i>
<i>sizuka-ni-te#ar-a-zu</i>	→ <i>sizuka-ni#ar-a-zu</i>
<i>sizuka-ni-te#ar-Ø-ru#NP</i>	→ <i>sizuka-ni#ar-Ø-ru#NP</i> / <i>sizuka-ni-te#ar-Ø-ru#NP</i>
<i>sizuka-ni-te#ar-Ø-re</i>	→ <i>sizuka-ni#ar-Ø-re</i> / <i>sizuka-ni-te#ar-Ø-re</i>

Likewise, in pseudo-copular forms such as *gakusei=n-a=no*, *gakusei=n-ar-a(-ba)* and *gakusei=ni#ar-a-zu* (lit.), the suffix *-te* does not surface in similar contexts. Thus, *-te* is assumed to be deleted obligatorily before *#ar-Ø-Ø=no*¹², *#ar-a-ba* and *#ar-a-zu*, as follows:

<i>gakusei=ni-te#ar-Ø-Ø=no</i>	→ <i>gakusei=ni#ar-Ø-Ø=no</i>
<i>gakusei=ni-te#ar-a-ba</i>	→ <i>gakusei=ni#ar-a-ba</i>
<i>gakusei=ni-te#ar-a-zu</i>	→ <i>gakusei=ni#ar-a-zu</i>

8. Amalgamation of *-ni* with *-te*, *-temo* and *-tatte*

The NA thematic suffix *-ni* and the postposition *=ni* amalgamate with the suffixes *-te*, *-temo*, *-tatte*, yielding *-de*, *-demo*, *-datte* and *=de*, *=demo*, *=datte*, respectively. Thus:

<i>sizuka-ni-te</i>	→ <i>sizuka-de</i>
<i>sizuka-ni-temo</i>	→ <i>sizuka-demo</i>

te#ar-Ø-Ø#wake=ga, *sizuka-ni-te#ar-Ø-Ø#bakari=ni* and *sizuka-ni-te#ar-Ø-Ø#mon=ni-te*. To reflect this in the SF computation, 'Delete *-te* following *-ni* before *#ar-Ø-Ø#NP*, *#ar-Ø-Ø=no*, *#ar-a-ba* and *#ar-a-zu*' in operation 7 should be revised to 'Delete *-te* following *-ni* before the following sequences: *#ar-Ø-Ø#NP*, *#ar-Ø-Ø=no*, *#ar-Ø-Ø#dake*, *#ar-Ø-Ø#wake*, *#ar-Ø-Ø#bakari*, *#ar-Ø-Ø#mon*, *#ar-a-ba* and *#ar-a-zu*'.

¹² As with the formal noun *=no*, the suffix *-te* is also deleted in *=ni-te#ar-Ø-Ø* before the nouns *#dake*, *#wake*, *#bacari* and *#mon*, as in *gakusei=n-a#dake#at-te*, *gakusei=n-a#wake=ga*, *gakusei=n-a#bacari=ni* and *gakusei=n-a#mon(o)=de*, derived from *gakusei=ni-te#ar-Ø-Ø#dake#ar-Ø-te*, *gakusei=ni-te#ar-Ø-Ø#wake=ga*, *gakusei=ni-te#ar-Ø-Ø#bakari=ni* and *gakusei=ni-te#ar-Ø-Ø#mon=ni-te*, and in *=ni-te#ar-Ø-ru* before the noun *#mono*, as in *gakusei=n-ar-u#mono=wa*, derived from *gakusei=ni-te#ar-Ø-ru#mono=wa*. To reflect this in the SF computation, 'Delete *-te* following *=ni* before the following sequences: *#ar-Ø-Ø=no*, *#ar-a-ba* and *#ar-a-zu*' in operation 7 should be revised to 'Delete *-te* following *=ni* before the following sequences: *#ar-Ø-Ø=no*, *#ar-Ø-Ø#dake*, *#ar-Ø-Ø#wake*, *#ar-Ø-Ø#bakari*, *#ar-Ø-Ø#mon*, *#ar-Ø-ru#mono*, *#ar-a-ba* and *#ar-a-zu*'.

<i>sizuka-ni-tatte</i> ¹³	→ <i>sizuka-datte</i>
<i>gakusei=ni-te</i>	→ <i>gakusei=de</i>
<i>gakusei=ni-temo</i>	→ <i>gakusei=demo</i>
<i>gakusei=ni-tatte</i>	→ <i>gakusei=datte</i>

This amalgamation, assumed here to be synchronic, is historically attributable to the diachronic development *ni-te* → *n-te* → *n-de* → *de*, probably involving the deletion of *-i-*, the postnasal voicing of the voiceless obstruent *-t-* after *n-*, a process typical of the Yamato lexical stratum, and the subsequent deletion of *n-*.

9. Deletion of *-i* in *-ni* before *#ar-*

In Contemporary Japanese, the vowel *-i* in the NA thematic suffix *-ni* is deleted immediately before the one-argument existential verb stem *ar-*. Thus:

<i>sizuka-ni#ar-Ø-Ø#NP</i>	→ <i>sizuka-n-ar-Ø-Ø#NP</i>
<i>sizuka-ni#ar-Ø-ru#NP</i>	→ <i>sizuka-n-ar-Ø-ru#NP</i>
<i>sizuka-ni#ar-Ø-Ø=no</i>	→ <i>sizuka-n-ar-Ø-Ø=no</i>
<i>sizuka-ni#ar-a-ba</i>	→ <i>sizuka-n-ar-a-ba</i>
<i>sizuka-ni#ar-a-zu</i>	→ <i>sizuka-n-ar-a-zu</i>
<i>sizuka-ni#ar-Ø-re</i>	→ <i>sizuka-n-ar-Ø-re</i> ¹⁴

The same vowel *-i* in the postposition *=ni* is also deleted except before *ar-a-zu*. Thus:

<i>gakusei=ni#ar-Ø-Ø=no</i>	→ <i>gakusei=n-ar-Ø-Ø=no</i>
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¹³ Although the derivation *-ni-tatte* → *-datte* is neither diachronically attested nor consistent with our native-speaker intuition, we nevertheless posit *sizuka-ni-tatte* as the UF of *sizuka-datte* on the basis of the homogeneity of the following processes (cf. Makino 2025¹: 10, 22–23):

<i>sizuka-ni-te</i>	→ <i>sizuka-ni</i> + <i>=wa</i> + <i>-te</i>	→ <i>sizuka-ni-te=wa#ar-te</i>	→ <i>sizuka-de=wa#at-te</i>
<i>sizuka-ni-temo</i>	→ <i>sizuka-ni</i> + <i>=wa</i> + <i>-temo</i>	→ <i>sizuka-ni-te=wa#ar-temo</i>	→ <i>sizuka-de=wa#at-temo</i>
<i>sizuka-ni-tatte</i>	→ <i>sizuka-ni</i> + <i>=wa</i> + <i>-tatte</i>	→ <i>sizuka-ni-te=wa#ar-tatte</i>	→ <i>sizuka-de=wa#at-tatte</i>

¹⁴ This classical imperative form of NAs remains in use today in a literary register, primarily with optative or benedictive force rather than as a directive, as in *yasuraka-n-ar-e* 'may you be at peace', *sukoyaka-n-ar-e* 'may you be in good health' and *saiwai-n-ar-e* 'may you be happy/blessed', etc.

gakusei=ni#ar-a-ba → *gakusei=n-ar-a-ba*

gakusei=ni#ar-a-zu → *gakusei=ni#ar-a-zu*

10. Deletion of the conditional suffix *-ba* following *-n-ar-a*, *=n-ar-a* and *-tar-a*

In Contemporary Japanese, the conditional suffix *-ba* is very often deleted after *-n-ar-a*, *=n-ar-a* and, especially, *-tar-a*. Thus:

sizuka-n-ar-a-ba → *sizuka-n-ar-a* / *sizuka-n-ar-a-ba*

gakusei=n-ar-a-ba → *gakusei=n-ar-a* / *gakusei=n-ar-a-ba*

tab-e-tar-a-ba → *tab-e-tar-a* / *tab-e-tar-a-ba*

11. Deletion of all remaining instances of *-Ø* and *Ø-*

Any remaining occurrences of *-Ø* and *Ø-* are deleted. In other words, except in the environments specified in operation 6 above, the V thematic suffix *-Ø* has no overt phonological realisation. Thus:

kes-Ø-ru → *kes-ru*

kes-Ø-ru-mai → *kes-ru-mai*

kes-Ø-tar-a-ba → *kes-tar-a-ba*

S-Ø-rar-e → *S-rar-e*

S-Ø-sas-e → *S-sas-e*

Ø-e-ru, Ø-u-ru → *e-ru, u-ru*

sizuka-n-ar-Ø-Ø → *sizuka-n-ar*

12. Insertion of *i* before the morpheme boundary /

Before the morpheme boundary /, *i* is inserted after any stem-final consonant. This process does not need to refer separately to *-tutu*, *-nagara*, *-gatera*, *-ta**. Those forms are analysed as occurring outside this same boundary. Accordingly, the relevant environment is not the initial consonant of the following suffix, but the boundary / itself. Thus:

<i>ut/</i>	→ <i>uti/</i>
<i>ut/-tutu</i>	→ <i>uti/-tutu</i>
<i>ut/-nagara</i>	→ <i>uti/-nagara</i>
<i>ut/-gatera</i>	→ <i>uti/-gatera</i>
<i>ut/-ta-ku</i>	→ <i>uti/-ta-ku</i>
<i>taka-ku#ar/-ta-ku</i>	→ <i>taka-ku#ari/-ta-ku</i>
<i>gakusei=de#ar/-nagara</i>	→ <i>gakusei=de#ari/-nagara</i>

The insertion of *i* therefore applies uniformly before /, regardless of whether further material follows outside that boundary. In this sense, the stem form seen before */-tutu*, */-nagara*, */-gatera* and */-ta** is the same as that seen before / alone.

The verbal system therefore distinguishes two suffix classes. The first class consists of suffixes such as *-te*, *-temo*, *-tatte*, *-tar**, which attach directly to the consonant-final stem and trigger immediate alternations such as voicing, assimilation and vowel insertion. The second class consists of forms such as *-tutu*, *-nagara*, *-gatera*, *-ta**, which do not interact directly with the stem-final consonant. Instead, they are introduced outside the morpheme boundary /. As a result, the stem first undergoes the same process that appears before / alone, namely insertion of *i* after a stem-final consonant. Accordingly, the opposition is not simply between two sets of consonant-initial suffixes, but rather between suffixes that attach directly to the stem and suffixes that are introduced only outside the boundary /.

13. Deletion of the boundary /

For notational simplicity, the symbol /, previously used to mark a morpheme boundary, is eliminated throughout. Before a suffix, the suffixation marker - already identifies the relevant morpheme boundary, so a sequence such as */-* constitutes redundant double marking of the same juncture. Nor is / required word-finally: the end of the word necessarily coincides with the end of the final morpheme and therefore already constitutes a morpheme boundary. Thus, neither */-* before a suffix nor word-final / contributes any additional structural information, and both can be represented more economically without /.

14. Deletion of suffix-initial consonants

Suffix-initial *s*, *r* and *y* are deleted after any stem-final or suffix-final consonant. Thus:

kes-ru → *kes-u*

kes-ru-mai → *kes-u-mai*

kes-reba → *kes-eba*

kes-tar-yō → *kes-tar-ō* (→ *kesi-tar-ō*)

kes-yō → *kes-ō*

S-rar-e → *S-ar-e*

S-sas-e → *S-as-e*

15. Allomorphy of *-re* to *-ro* and *-yo* to *-i*

The imperative suffix *-re* is realised as *-ro* throughout¹⁵, and the other imperative suffix *-yo* is realised as *-i* only in the *k-irregular* verb. Thus:

tab-e-re → *tab-e-ro*

K-o-yo → *K-o-i*

16. Coalescence of *ow* into *ō* before the *t*-series suffixes

Before the suffixes *-te*, *-temo*, *-tatte* and *-tar**, the sequence *ow* becomes *ō*, obligatorily in *to-u* 'ask (a question)', underlyingly *tow-Ø-ru*, and *ko-u* 'beg', underlyingly *kow-Ø-ru*, and optionally in *iko-u* 'rest, relax', underlyingly *ikow-Ø-ru*. Thus:

tow-te → *tō-te*

kow-tar-a(-ba) → *kō-tar-a(-ba)*

¹⁵ In certain Hokkaidō varieties, this imperative suffix is realized as *-re* as in *tab-e-re*, which corresponds to the standard *tab-e-ro*.

17. Assimilation and insertion before the *t*-series suffixes

Before *-te*, *-temo*, *-tatte* and *-tar**, stem-final *r* and *w* assimilate to *t*. Thus:

-rt-, *-wt-* → *-tt-*, *-tt-*

Before the same suffixes, *s* and *k* surface as *si* and *ki* rather than assimilating to *t*. Accordingly:

-st-, *-kt-* → *-sit-*, *-kit-* → *-sit-*, *-it-*

This is because the assimilation *-st-*, *-kt-* → *-tt-*, *-tt-* is not permitted, except in the verb *ik-u* 'go', underlyingly *ik-Ø-ru*, where *-kt-* → *-tt-* is observed. Instead, *i* is inserted after *s* and *k*, yielding the sequences *si* and *ki*, after which *k* is deleted. The overall pattern may be stated as follows:

-tt-, *-st-*, *-rt-*, *-kt-*, *-wt-*
→ *-tt-*, *-st-*, *-tt-*, *-kt-*, *-tt-*
→ *-tt-*, *-sit-*, *-tt-*, *-kit-*, *-tt-*
→ *-tt-*, *-sit-*, *-tt-*, *-it-*, *-tt-*

Accordingly:

ut-te → *ut-te*
kes-te → *kesi-te*
kir-te → *kit-te*
kak-te → *kaki-te* → *kai-te*
ik-te → *it-te*
kaw-te → *kat-te*
ow-te → *ot-te*

18. Voicing in the *t*-series suffixes

In the suffixes *-te*, *-temo*, *-tatte* and *-tar**, suffix-initial *t* becomes *d* after stem-final *b*, *g*, *m*, *n*, that is, after consonants specified as [–sonorant, +voiced] or [+nasal]. Thus:

-bt-, -gt-, -mt-, -nt-
 → -bd-, -gd-, -md-, -nd-

These suffixes therefore form a class that attaches directly to a consonant-final stem and triggers immediate morphophonological interaction with it. Thus:

nug-te → *nug-de*
sin-te → *sin-de*
tob-te → *tob-de*
yom-te → *yom-de*

19. Alternations before the *d*-series allomorphs of the *t*-series suffixes

Before the *d*-series allomorphs *-de*, *-demo*, *-datte* and *-dar** of the *t*-series suffixes, stem-final *b* and *m* become *n*. This alternation may be understood as the final phase in a process involving nasalisation followed by place assimilation. Before the same allomorphs, by contrast, *g* surfaces as *i*. More precisely, this development is analysed not as a direct change *g* → *i*, but as a process involving insertion of *i* followed by deletion of *g*. The sequence may be represented as follows:

-bt-, -mt-, -nt-, -gt-
 → -bd-, -md-, -nd-, -gd-
 → -md-, -md-, -nd-, -gd-
 → -nd-, -nd-, -nd-, -gd-
 → -nd-, -nd-, -nd-, -gid-
 → -nd-, -nd-, -nd-, -id-

The interpretation is as follows. First, *t* becomes voiced after a nasal or a voiced obstruent. Next, *b* becomes *m* in order to avoid a sequence of voiced obstruents, as in *-bd-* → *-md-*. Then, *m* becomes *n* by place assimilation to *d*, as in *-md-* → *-nd-*. In the case of *g*, nasalisation to *ŋ*, as in *-gd-* → *-ŋd-*, is excluded — possibly by structure preservation — and assimilation to *d*, as in *-gd-* → *-dd-*, is likewise excluded, since sequences of identical voiced obstruents are prohibited in the Yamato and Sino-Japanese lexical strata of Japanese. The cluster is therefore broken up by

insertion of the default epenthetic vowel *i*, after which *g* is deleted, as in *-gd-* → *-gid-* → *-id-*.
Thus:

*to**b**-de* → *to**m**-de* → *to**n**-de*

*yo**m**-de* → *yo**n**-de*

nug-de → *nug**i**-de* → *nui-de*

20. Deletion of *w* except before *a**

The consonant *w* is deleted except before *a**. Thus:

*ka**w**-u* → *ka-u*

*ka**w**-ō* → *ka-ō*

*to**w**-i* → *to-i*

*to**w**-eba* → *to-eba*

By contrast, *w* is retained before *a**. Thus:

*ka**w**-a-na-ku* → *ka**w**-a-na-ku*

*to**w**-ar-e* → *to**w**-ar-e*

This is enforced by the exclusion of Yamato and Sino-Japanese **wi*, **we*, **wo*, **wu* sequences.

21. Vowel deletion in *-ku#at-* and *-ku#ar-*, yielding *-k+at-* and *-k+ar-*

Before *-ta**, *-ku#at* changes to *-k+at*. Before *-ō*, *-e* and *-a-zu*, *-ku#ar* changes to *-k+ar*. Optionally, before *-u-mai*, *-ku#ar* changes to *-k+ar* in a more literary register. Thus:

taka-ku#at-ta → *taka-k+at-ta*

taka-ku#at-tar-ō → *taka-k+at-tar-ō*

taka-ku#at-tar-a(-ba) → *taka-k+at-tar-a(-ba)*

taka-ku#at-tar-i → *taka-k+at-tar-i*

taka-ku#ar-ō → *taka-k+ar-ō*

taka-ku#ar-e → *taka-k+ar-e*

taka-ku#ar-a-zu → *taka-k+ar-a-zu*

taka-ku#ar-u-mai → *taka-k+ar-u-mai* / *taka-ku#ar-u-mai*

This vowel deletion may be taken to indicate that the VA stem in *-ku* and the light verb stem *ar-* have come to form a single compound stem, as stated in Section 3 of the main text, though *taka-k+ar-ō*, *taka-k+ar-e*, *taka-k+ar-a-zu* and *taka-k+ar-u-mai* are either more or less literal forms or usually occur in fixed expressions such as *ō-k+ar-ō=ga sukuna-k+ar-ō=ga* 'whether many or few' and *ō-k+ar-e sukuna-k+ar-e* 'more or less', from the VAs *ōi* 'be many/much' and *sukunai* 'be few/little'. Note that besides the contracted forms such as *utukusi-k+ar-ō* from the VA *utukusii* 'be beautiful', which mainly mark assumption, the non-contracted forms such as *utukusi-ku#ar-ō*, which mark volition or deonticity, are attested. It may be assumed that, in the former, *ar-* contracted with VA stems ending in *-ku* functions as a light verb introduced to enable purely verbal suffixation by *-yō*, *-re*, *-zu* and *-mai*, whereas, in the latter, the non-contracted *ar-* is likely to be analysed as a one-argument existential predicate.

22. Vowel deletion in the suffix–suffix amalgam *-de* and the postposition–suffix amalgam *=de*

In nominal-adjectival and pseudo-copular non-past forms such as *sizuka-d-a* and *gakusei=d-a*, the deletion of *-e* in *-de* and *=de* is obligatory when the non-past tense suffix *-ru* has alternated with $\emptyset$. In some other NA and pseudo-copular forms, on the other hand, this deletion is facultative in a somewhat less formal register, as in *sizuka-d-ar-ō*, *gakusei=d-ar-ō*, *sizuka-d-at-ta* and *gakusei=d-at-ta*, paired with the more formal *sizuka-de#ar-ō*, *gakusei=de#ar-ō*, *sizuka-de#at-ta* and *gakusei=de#at-ta*. Thus:

sizuka-de#ar → *sizuka-d-ar*

sizuka-de#ar-ō → *sizuka-d-ar-ō* / *sizuka-de#ar-ō*

sizuka-de#at-tar → *sizuka-d-at-tar* / *sizuka-de#at-tar*

gakusei=de#ar → *gakusei=d-ar*

gakusei=de#ar-ō → *gakusei=d-ar-ō* / *gakusei=de#ar-ō*

gakusei=de#at-tar → *gakusei=d-at-tar* / *gakusei=de#at-tar*

23. Deletion of word-final *r*

Word-final *r* is deleted. Thus:

<i>kesi-tar</i>	→ <i>kesi-ta</i>
<i>sizuka-d-ar</i>	→ <i>sizuka-d-a</i>
<i>gakusei=d-at-tar</i>	→ <i>gakusei=d-at-ta</i>

This operation eliminates a consonant that is not permitted word-finally at the surface.

24. Deletion of *ar-a-* before *-na**

Before the negative suffix *-na**, the verbal stem *ar-a-* is deleted. Thus:

<i>ar-a-na-i</i>	→ <i>na-i</i>
<i>taka-ku#ar-a-na-ke-reba</i>	→ <i>taka-ku#na-ke-reba</i>
<i>sizuka-de#a-ra-na-ku</i>	→ <i>sizuka-de#na-ku</i> (→ <i>sizuka-de=wa#na-ku</i>)

This indicates, as stated in Section 7 of the main text, the obligatory suppletion in which the negative forms of the verb *aru* is replaced by the inflected forms of the VA *nai* in Contemporary Japanese.

25. Insertion of *=wa* in nominal-adjectival and pseudo-copular negative forms

In nominal-adjectival and pseudo-copular negative forms in *-de* and *=de*, respectively, partial negation with the scope-limiting *=wa* is the default pattern, as discussed in Makino (2025¹: 10–19; 2025²: 21–24). Thus:

<i>sizuka-de#na-i</i>	→ <i>sizuka-de=wa#na-i</i>
<i>sizuka-de#ar-u-mai</i>	→ <i>sizuka-de=wa#ar-u-mai</i>
<i>gakusei=de#na-i</i>	→ <i>gakusei=de=wa#na-i</i>
<i>gakusei=de#ar-u-mai</i>	→ <i>gakusei=de=wa#ar-u-mai</i>

26. Contraction of *-de=wa* and *=de=wa* into *-dy-a* and *=dy-a*

In spoken Standard Japanese, the sequences *-de=wa* and *=de=wa* in nominal-adjectival and pseudo-copular negative forms are frequently contracted into *-dy-a* and *=dy-a*, respectively. Thus:

sizuka-de=wa#na-i → *sizuka-dy-a#na-i*

sizuka-de=wa#ar-u-mai → *sizuka-dy-a#ar-u-mai*

gakusei-de=wa#na-i → *gakusei-dy-a#na-i*

gakusei-de=wa#ar-u-mai → *gakusei-dy-a#ar-u-mai*

To the best of our knowledge, no previous analysis appears to have presented a single partially ordered system of serial and mutually independent operations capable of deriving, from explicit underlying representations, the basic inflected forms of Japanese true verbs, verbal adjectives, nominal adjectives and pseudo-copular constructions.